

Fast On-device LLM Inference with NPUs

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Abstract

On-device inference for Large Language Models (LLMs), driven by increasing privacy concerns and advancements of mobile-sized models, has gained significant interest. However, even mobile-sized LLMs (e.g., Gemma-2B) encounter unacceptably high inference latency, often bottlenecked by the prefill stage in tasks like screen UI understanding.

We present 11m.npu, the first LLM inference system utilizing on-device Neural Processing Unit (NPU) offloading to reduce prefill latency. 11m.npu enhances NPU offloading efficiency by re-constructing the prompt and model in three levels: (1) At *prompt level*, it divides variable-length prompts into multiple fixed-sized chunks while maintaining data dependencies; (2) At *tensor level*, it identifies and extracts significant outliers to run on the CPU/GPU in parallel with minimal overhead; (3) At *block level*, it schedules Transformer blocks in an out-of-order manner to the CPU/GPU and NPU based on their hardware affinity and sensitivity to accuracy. Compared to competitive baselines, 11m.npu achieves 22.4× faster prefill speed and 30.7× energy savings

on average, and up to 32.8× speedup in an end-to-end real-world application. For the first time, 11m.npu achieves more than 1,000 tokens/sec prefilling for a billion-sized model.

CCS Concepts: • Human-centered computing → Ubiquitous and mobile computing; • Computer systems organization → Heterogeneous (hybrid) systems; • Computing methodologies → Natural language processing.

Keywords: Mobile computing, large language model, NPU

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1 Introduction

With rising privacy concerns [5], there is growing interest in running Large Language Models (LLMs) locally on mobile devices, known as *on-device LLMs* [99, 100, 102], e.g., Apple Intelligence [2] and Android AI Core [1]. Meanwhile, the advancement of mobile-sized language models (1B–10B parameters), such as Qwen2-1.5B and Phi3-3.7B, has demonstrated their ability to perform comparably to significantly larger models like GPT-3, despite having a reduced parameter count [24, 26, 27, 60]. This progress makes the deployment of on-device language models feasible. Without giving away private data, on-device LLM inference catalyzes novel mobile applications [90], such as UI task automation [55] (e.g., translating users’ language commands into UI operations such as “forward the unread emails to Alice”) and automated message reply [20].

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通过 NPUs 进行快速设备上 LLM 推理

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抽象的

在日益增长的隐私问题和移动规模模型的进步的推动下，大型语言模型 (LLM) 的设备端推理引起了人们的极大兴趣。然而，即使是移动大小的 LLM（例如 Gemma-2B）也会遇到不可接受的高推理延迟，通常会受到屏幕 UI 理解等任务中预填充阶段的瓶颈。

我们推出了 llm.npu，这是第一个利用设备上神经处理单元 (NPU) 卸载来减少预填充延迟的 LLM 推理系统。llm.npu 通过在三个层面上重构提示和模型来提高 NPU 卸载效率：（1）在提示层面，它将可变长度提示划分为多个固定大小的块，同时保持数据依赖性；（2）在张量级别，它识别并提取显著的异常值，以最小的开销在 CPU/GPU 上并行运行；（3）在块级别，它根据硬件亲和性和对准确性的敏感性，以无序方式将 Transformer 块调度到 CPU/GPU 和 NPU。与竞争基准相比，llm.npu 的预填充速度提高了 22.4×，能源节省了 30.7×

平均而言，在端到端的实际应用中加速高达 32.8×。llm.npu 首次实现了十亿规模模型每秒超过 1,000 个代币的预填充。

CCS 概念：• 以人为本的计算 → 无处不在的移动计算；
• 计算机系统组织 → 异构（混合）系统；
方法 → 自然语言处理。

关键词：移动计算、大语言模型、NPU

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1 简介

随着隐私问题的日益严重[5]，人们对在移动设备上本地运行大型语言模型 (LLM) 越来越感兴趣，称为设备上 LLM [99,100,102]，例如 Apple Intelligence [2] 和 Android AI Core [1]。与此同时，移动大小的语言模型（1B-10B 参数）的进步，例如 Qwen2-1.5B 和 Phi3-3.7B，已经证明了它们的性能与 GPT-3 等更大的模型相当，尽管参数数量减少了 [24,26,27,60]。这一进展使得设备上语言模型的部署变得可行。在不泄露私人数据的情况下，设备上的 LLM 推理可以催化新颖的移动应用程序 [90]，例如 UI 任务自动化 [55]（例如，将用户的语言命令翻译为 UI 操作，例如“将未读电子邮件转发给 Alice”）和自动消息回复 [20]。

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However, the high inference latency remains as a significant obstacle to practical on-device LLMs. To accomplish a UI task, LLM needs to ingest the screen view hierarchy (typically 600–800 tokens [22, 82]) to generate corresponding UI operations step by step [81]. As will be shown in §2.1, each such step takes 8.1 seconds for Qwen1.5-1.8B [27], and thereby more than 40 seconds to finish a 5-step UI task. Similarly, the Gemma-2B model requires 26.7 seconds to automatically reply to an email by mimicking the user’s tone based on historical email data (with 1500 tokens). Diving into those tasks, we find out that the prompt processing (prefill stage) often dominates the end-to-end inference latency, e.g., 94.4%–98.8% for UI automation tasks. This is because on-device LLM tasks often involve long-context understanding for handle personalized tasks. Unfortunately, existing research efforts primarily focus on accelerating the text generation speed (decoding stage), such as activation sparsity [74, 94] and speculative decoding [50, 61, 95]. Therefore, this work mainly targets improving the prefill speed of on-device LLMs.

LLM prefilling is compute-bounded [69, 83, 106]; yet, mobile CPU and GPU have limited parallel computing capacity [40, 97]. Instead, we are motivated by a key opportunity that Neural Processing Units (NPUs) are ubiquitously available in modern mobile devices, e.g., Qualcomm Hexagon NPU and Google Edge TPU. These mobile NPUs are efficient at integer vector operations, delivering computing capability up to 73 TOPS [17]. On CNNs, their improvements over mobile CPU/GPUs are demonstrated to be up to 18×/4×, respectively [86]. Mobile NPUs are also more energy-efficient, and have less workloads contention as compared to CPU/GPU.

Surprisingly, with such promised advantages, there exists no systems supporting LLM inference on COTS mobile NPUs. Indeed, our preliminary efforts show that directly employing mobile NPUs for LLM inference does not offer performance benefits due to the following challenges.

- *Costly preparation for variable-length prompts.* Mobile NPUs typically support only inference on static shapes, while LLM prompt length is dynamic (with a max context length). Re-preparing and optimizing the LLM execution graph on NPUs for each different-sized prompt is costly on mobile devices (e.g., 11 seconds for the Gemma-2B model). On the other hand, simply padding the LLM request to the same max context length wastes precious computing resources.

- *Mismatch between LLM quantization algorithms and mobile NPU design.* Due to the presence of outlier activations [33, 84], state-of-the-art LLM quantization methods often use per-group quantization to maintain high accuracy. It partitions original activations and weights tensors into multiple groups with independent quantization scales to avoid the impacts of outliers on others. However, our investigation shows that mobile NPUs cannot perform per-group MatMul directly (Table 2). Instead, they must split the MatMul into multiple group-sized sub-tensor MatMuls and then reduce the sub-tensor intermediate results using a float sum

operation. This process hampers NPU efficiency and incurs up to 10.7× performance overhead.

- *Floating point (FP) operations cannot be eliminated.* Mobile NPUs generally provide significant integer-based MatMul acceleration but are weak at FP operations. However, LLMs can be hardly quantized into integer-only execution with minimal accuracy loss. Existing quantized LLMs still rely on float operators like LayerNorm and Attention. Scheduling those FP operators out of NPU easily increases the inference critical path.

This work presents 11m.npu, the first LLM inference system with efficient on-device NPU offloading. The primary design goal of 11m.npu is to reduce the prefill latency and energy consumption. It targets the mainstream decoder-only transformer architecture of LLMs (e.g., LLaMA, GPT, etc). The key idea is to maximize prefill execution on mobile NPUs to accelerate integer computation while keeping essential float operations on the CPU/GPU to maintain accuracy. To overcome the aforementioned challenges and enhance NPU offloading efficiency, 11m.npu re-constructs the prompt and model at three levels: (1) *At prompt level:* 11m.npu divides variable-length prompts into multiple fixed-sized chunks while maintaining data dependencies; (2) *At tensor level:* 11m.npu identifies and extracts significant outliers to run on the CPU/GPU; (3) *At block level:* 11m.npu schedules Transformer blocks to the CPU/GPU and NPU based on their hardware affinity and sensitivity to accuracy. The corresponding novel techniques are detailed as follows:

- *Chunk-sharing graphs (§3.2)* 11m.npu splits variable-length prompts into fixed-sized “chunks” with pre-built sub-graphs, reducing graph preparation time. Each variable-length prompt is then executed on those graphs with chunk-level causal dependency. This approach leverages the insight that the token generation depends only on preceding tokens in the decoder-only LLMs. However, loading multiple pre-built chunk graphs simultaneously incurs significant memory overhead, e.g., 2–4× more than the LLM weights. To address this, 11m.npu further identifies operators that are irrelevant to prompt size (e.g., FFN) and shares them across each chunk execution, reducing the memory overhead by up to 4×.

- *Shadow outlier execution (§3.3)* addresses the activation outlier problem without compromising NPU efficiency. Without changing per-tensor MatMul on NPU, 11m.npu extracts the activation outlier channels into a compact tensor and executes it on the CPU/GPU in parallel, since outliers are very sparse (0.1%–0.3% of total channels). This design further raises issues of increased memory footprint due to duplicating MatMul weights in CPU memory and synchronization overhead between CPU/GPU and NPU. Therefore, based on the observation that outliers are more likely to appear at a small set of channel positions, 11m.npu optimizes memory usage by only keeping those “hot channels” weights in memory, and retrieves others from disk on demand. 11m.npu

然而，高推理延迟仍然是实际设备上 LLM 的重大障碍。为了完成 UI 任务，LLM 需要摄取屏幕视图层次结构（通常为 600-800 个标记 [22, 82]）以逐步生成相应的 UI 操作 [81]。正如 §2.1 中所示，对于 Qwen1.5-1.8B [27]，每个此类步骤需要 8.1 秒，因此完成 5 步 UI 任务需要 40 秒以上。同样，Gemma-2B 模型根据历史电子邮件数据（具有 1500 个令牌）模仿用户的语气，需要 26.7 秒来自动回复电子邮件。深入研究这些任务，我们发现提示处理（预填充阶段）通常主导端到端推理延迟，例如 UI 自动化任务的 94.4% - 98.8%。这是因为设备上的 LLM 任务通常涉及长上下文理解以处理个性化任务。不幸的是，现有的研究工作主要集中在加快文本生成速度（解码阶段），例如激活稀疏性 [74, 94] 和推测解码 [50, 61, 95]。因此，这项工作的主要目标是提高设备上 LLM 的预填充速度。

LLM 预填充受计算限制 [69, 83, 106]；然而，移动 CPU 和 GPU 的并行计算能力有限 [40, 97]。相反，我们受到神经处理单元（NPU）在现代移动设备中无处不在的关键机遇的激励，例如高通 Hexagon NPU 和 Google Edge TPU。这些移动 NPU 在整数向量运算方面非常高效，计算能力高达 73 TOPS [17]。在 CNN 上，它们相对于移动 CPU/GPU 的改进被证明分别高达 $18\times/4\times$ [86]。与 CPU/GPU 相比，移动 NPU 也更节能，并且工作负载争用更少。

令人惊讶的是，尽管有这些承诺的优势，但目前还没有支持 COTS 移动 NPU 上的 LLM 推理的系统。事实上，我们的初步工作表明，由于以下挑战，直接使用移动 NPU 进行 LLM 推理并不能提供性能优势。

- 可变长度提示的准备成本高昂。移动 NPU 通常仅支持静态形状的推理，而 LLM 提示长度是动态的（具有最大上下文长度）。在移动设备上，为每个不同大小的提示在 NPU 上重新准备和优化 LLM 执行图的成本很高（例如，Gemma-2B 模型需要 11 秒）。另一方面，简单地将 LLM 请求填充到相同的最大上下文长度会浪费宝贵的计算资源。

- LLM 量化算法与移动 NPU 设计之间的不匹配。由于异常激活的存在 [33, 84]，最先进的 LLM 量化方法通常使用每组量化来保持高精度。它将原始激活和权重张量划分为多个具有独立量化尺度的组，以避免异常值对其他组的影响。然而，我们的研究表明移动 NPU 无法直接执行每组 MatMul（表 2）。相反，他们必须将 MatMul 拆分为多个组大小的子张量 MatMuls，然后使用浮点和来减少子张量中间结果

手术。此过程会影响 NPU 效率并产生高达 $10.7\times$ 的性能开销。

- 浮点 (FP) 运算无法消除。移动 NPU 通常提供显著的基于整数的 Mat-Mul 加速，但在 FP 运算方面较弱。然而，LLM 很难以最小的精度损失量化为纯整数执行。现有的量化 LLM 仍然依赖于像 LayerNorm 和 Attention 这样的浮点运算符。将这些 FP 算子调度到 NPU 之外可以轻松地增加推理关键路径。

这项工作提出了 llm.npu，这是第一个具有高效设备上 NPU 卸载功能的 LLM 推理系统。llm.npu 的主要设计目标是减少预填充延迟和能耗。它针对的是 LLM 的主流仅解码器 Transformer 架构（例如 LLaMA、GPT 等）。关键思想是最大限度地提高移动 NPU 上的预填充执行速度，以加速整数计算，同时在 CPU/GPU 上保留必要的浮点运算以保持准确性。为了克服上述挑战并提高 NPU 卸载效率，llm.npu 在三个层面上重新构建了提示和模型：（1）在提示级别：llm.npu 将可变长度提示划分为多个固定大小的块，同时保持数据依赖性；（2）在张量级别：llm.npu 识别并提取显著异常值以在 CPU/GPU 上运行；（3）在块级别：llm.npu 根据硬件亲和性和对准确性的敏感性将 Transformer 块调度到 CPU/GPU 和 NPU。相应的新技术详述如下：

- 块共享图 (§3.2) llm.npu 将可变长度提示拆分为带有预先构建的子图的固定大小的“块”，从而减少了图准备时间。然后，在具有块级因果依赖性的那些图上执行每个可变长度提示。这种方法利用了令牌生成仅依赖于仅解码器 LLM 中的先前令牌的见解。然而，同时加载多个预构建的块图会产生大量内存开销，例如比 LLM 权重多 $2-4\times$ 。为了解决这个问题，llm.npu 进一步识别与提示大小无关的运算符（例如 FFN），并在每个块执行中共享它们，从而将内存开销减少多达 $4\times$ 。

- 影子离群值执行 (§3.3) 在不影响 NPU 效率的情况下解决了激活离群值问题。在不更改 NPU 上每个张量 MatMul 的情况下，llm.npu 将激活离群值通道提取到紧凑张量中，并在 CPU/GPU 上并行执行，因为离群值非常稀疏（占总通道的 0.1%–0.3%）。由于 CPU 内存中重复的 MatMul 权重以及 CPU/GPU 和 NPU 之间的同步开销，该设计进一步引发了内存占用增加的问题。因此，基于异常值更有可能出现在一小部分通道位置的观察结果，llm.npu 通过仅将那些“热通道”权重保留在内存中来优化内存使用，并根据需要从磁盘检索其他权重。LLM.npu

also prunes unimportant outliers at layer-level by measuring outliers' importance to reduce the synchronization overhead.

- *Out-of-order subgraph execution* (§3.4) `llm.npu` is guided by a key insight that multiple subgraphs can be scheduled in an out-of-order manner, without strictly following the chunk sequence in the original prompt. This significantly enlarges the scheduling space of `llm.npu` to minimize the execution bubbles brought by CPU/GPU float operation. Given that finding the optimal out-of-order execution order is an NP-hard problem, `llm.npu` employs a microsecond-level online scheduling algorithm. This algorithm is based on the observation that the workload of the NPU is heavier and constitutes the critical path. Therefore, when selecting which subgraph to execute, `llm.npu` prioritizes those subgraphs having a more significant impact on reducing NPU stalls, rather than focusing solely on the execution latency of the subgraph. Notably, `llm.npu`'s scheduling algorithm does not maximize parallel processing capability. Instead, `llm.npu` aims to maximize the utilization of NPUs while minimizing the impact of CPU/GPU workloads.

Implementation and evaluations. We have implemented `llm.npu` on top of MLLM [23] and QNN [25], with 10K lines of C/C++ and assembly code. We evaluated `llm.npu` with five mobile-sized LLMs (Qwen1.5-1.8B [27], Gemma-2B [9], phi2-2.7B [16], LLaMA-2-7B [11], and Mistral-7B [14]), four LLM benchmarks, and two mobile devices (Xiaomi 14 and Redmi K60 Pro). We compared `llm.npu` with five competitive baselines, including three industrial open-source engines (llama.cpp [59], TFLite [57], MNN [49]) and two state-of-the-art research prototypes (MLC-LLM [75] and PowerInfer-v2 [94]). The experiments show that `llm.npu` significantly and consistently outperforms all baselines in terms of prefill latency and energy consumption while preserving inference accuracy (<1% loss compared to FP16). It is 7.3×–18.4× faster than baselines on CPU, and 1.3×–43.6× on GPU with a prompt length of 1024. It also achieves a 1.9×–59.5× energy reduction. To our best knowledge, `llm.npu` is the first system that achieves >1000 tokens/sec of prefill speed on COTS mobile devices for billion-sized LLMs. In end-to-end real-world applications, `llm.npu` reduces the inference latency (prefill+decode) by 1.4×–32.8× compared to the baselines.

Contributions are summarized as follows:

- We thoroughly investigate the challenges and opportunities of accelerating LLM prefilling with mobile NPUs.
- We present the first LLM inference engine with efficient mobile NPU offloading, featuring three novel techniques: chunk-sharing graph, shadow outlier execution, and out-of-order subgraph execution.
- We perform comprehensive experiments on `llm.npu` that demonstrate its superior performance over competitive baselines. The code of `llm.npu` is available at <https://github.com/UbiquitousLearning/mlm>.

2 Background

2.1 On-device LLM Inference Analysis

On-device LLMs are increasingly used in cutting-edge scenarios such as Apple Intelligence [2], UI automation [81], and automated email reply [20], due to the enhanced privacy protections. To empower these applications, many lightweight LLMs have been developed, as summarized in Table 1. For instance, the Qwen1.5-1.8B model on the llama.cpp exhibits delays of 8.1 seconds for one-step UI automation, 21.7 seconds for automated email replies, and 8.4 seconds for chat summary on average using mobile CPUs, as evaluated on the DroidTask [22, 82], LongBench datasets [28] and Persona-Chat [48], which is impractical for real-world deployment. However, their inference latency remains a significant challenge.

To explore this issue in detail, we break down the inference procedure discussed above. The results are summarized in Figure 1. Surprisingly, we find that compared to the decoding phase that has been extensively studied, the prefill stage is often the bottleneck in typical mobile applications. For example, the prefill stage constitutes 88.3%–98.8% of the total latency on a mobile CPU for UI task automation, chat summaries, and context-aware generation tasks. Even when using mobile GPUs, the prefill phase remains predominant, accounting for 54.2% to 91.7% of the total latency. As the prompt length increases, the prefill stage's proportion of the total inference time also rises.

Several factors contribute to this situation: (1) Mobile CPUs/GPUs lack the parallelism capabilities of cloud GPUs [40, 97], being primarily designed for handling application logic or rendering tasks. (2) Mobile LLM tasks often require long prompts for personalized, context-aware generation. For instance, automated email reply often requires extensive user data in prompt, such as historical emails, schedules, and location information, amounting to 1,168–1,835 tokens in the LongBench dataset [28]. Similarly, on-device LLMs engaged in UI automation must process a substantial volume of UI annotation tokens (XML or HTML) and user commands, ranging from 505–827 tokens in the DroidTask dataset [22, 82]. Yet, the output lengths for these applications are typically short, averaging 7.9 tokens for the LongBench dataset and 3.5 tokens for the DroidTask dataset. In the context of chat summaries, such as the Persona-Chat dataset [48], the prompt and output lengths are relatively balanced, with input sizes between 488–584 tokens and an average output of 44 tokens. (3) Mobile LLMs now support long context windows. For instance, recent models like Qwen2-1.5B can accommodate context windows of up to 32K tokens, as illustrated in Table 1.

2.2 Opportunity: Mobile NPUs

To optimize prefill latency, `llm.npu` leverages a key opportunity: modern mobile SoCs ubiquitously include mobile

还通过测量异常值的重要性来修剪层级别的不重要异常值，以减少同步开销。

- 无序子图执行 (§3.4) `llm.npu` 以一个关键见解为指导，即可以以无序方式调度多个子图，而无需严格遵循原始提示中的块顺序。这显着扩大了 `llm.npu` 的调度空间，以最大限度地减少 CPU/GPU 浮动操作带来的执行气泡。鉴于寻找最佳乱序执行顺序是一个 NP 难题，`llm.npu` 采用了微秒级在线调度算法。该算法基于观察到 NPU 的工作负载较重并且构成关键路径。因此，在选择执行哪个子图时，`llm.npu` 会优先考虑那些对减少 NPU 停顿影响更显著的子图，而不是仅仅关注子图的执行延迟。值得注意的是，`llm.npu` 的调度算法并没有最大化并行处理能力。相反，`llm.npu` 的目标是最大限度地提高 NPU 的利用率，同时最大限度地减少 CPU/GPU 工作负载的影响。

实施和评估。我们在 MLLM [23] 和 QNN [25] 之上实现了 `llm.npu`，其中包含 10K 行 C/C++ 和汇编代码。我们使用五个移动大小的 LLM (Qwen1.5-1.8B [27]、Gemma-2B [9]、phi2-2.7B [16]、Llama-2-7B [11] 和 Mistral-7B [14])、四个 LLM 基准测试和两个移动设备 (小米 14 和 Redmi K60 Pro) 评估了 `llm.npu`。我们将 `llm.npu` 与五个竞争基准进行了比较，包括三个工业开源引擎 (llama.cpp [59]、TFLite [57]、MNN [49]) 和两个最先进的研究原型 (MLC-LLM [75] 和 PowerInfer-v2 [94])。实验表明，`llm.npu` 在预填充延迟和能耗方面显着且一致地优于所有基线，同时保持推理精度 (与 FP16 相比 < 损失 1%)。它比 CPU 上的基准快 7.3x–18.4x，在 GPU 上比基准快 1.3x–43.6x，提示长度为 1024。它还实现了 1.9x–59.5x 能耗降低。据我们所知，`llm.npu` 是第一个在 CO TS 移动设备上为数亿规模的 LLM 实现 >1000 个令牌/秒预填充速度的系统。在端到端的实际应用中，与基线相比，`llm.npu` 将推理延迟 (预填充+解码) 减少了 1.4x–32.8x。

贡献总结如下：

- 我们深入研究了使用移动 NPU 加速 LLM 预填充的挑战和机遇。
- 我们提出了第一个具有高效移动 NPU 卸载功能的 LLM 推理引擎，具有三种新颖的技术：块共享图、影子异常值执行和乱序子图执行。
- 我们在 `llm.npu` 上进行了全面的实验，证明了其优于竞争基线的性能。`llm.npu` 的代码可在 <https://github.com/UbiquitousLearning/mlm> 获取。

2 背景

2.1 设备上 LLM 推理分析

由于增强的隐私保护，设备上的 **大语言模型** 越来越多地用于尖端场景，例如 Apple Intelligence [2]、UI 自动化 [81] 和自动电子邮件回复 [20]。为了支持这些应用程序，人们开发了许多轻量级 LLM，如表 1 所示。例如，根据 DroidTask [22, 82]、LongBench 数据集的评估，llama.cpp 上的 Qwen1.5-1.8B 模型在使用移动 CPU 时，单步 UI 自动化延迟为 8.1 秒，自动电子邮件回复延迟为 21.7 秒，聊天摘要延迟为 8.4 秒。[28] 和 Persona-Chat [48]，这对于现实世界的部署来说是不切实际的。然而，它们的推理延迟仍然是一个重大挑战。

为了详细探讨这个问题，我们分解了上面讨论的推理过程。结果如图 1 所示。令人惊讶的是，我们发现与已广泛研究的解码阶段相比，预填充阶段通常是典型移动应用中的瓶颈。例如，预填充阶段占 UI 任务自动化、聊天摘要和上下文感知生成任务的移动 CPU 总延迟的 8.3%–98.8%。即使使用移动 GPU，预填充阶段仍然占主导地位，占总延迟的 54.2% 至 91.7%。随着提示长度的增加，预填充阶段占总推理时间的比例也会增加。

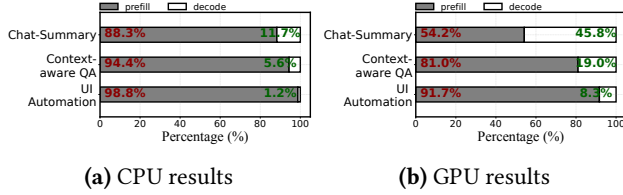
造成这种情况的因素有几个：(1) 移动 CPU/GPU 缺乏云 GPU 的并行能力 [40]，(2) 主要设计用于处理应用程序逻辑或渲染任务，(3) 移动 **大语言模型** 通常需要很长的提示才能进行个性化、上下文感知的生成。例如，自动电子邮件回复通常需要大量的用户数据，例如历史电子邮件、日程安排和位置信息，在 LongBench 数据集中总计有 1,168–1,835 个令牌 [28]。同样，从事 UI 自动化的设备上 **大语言模型** 处理大量的 UI 注释标记 (XML 或 HTML) 和用户命令，在 DroidTask 数据集中有 505–827 个标记 [22, 82]。然而，这些应用程序的输出长度通常很短，LongBench 数据集平均为 7.9 个标记，DroidTask 数据集平均为 3.5 个标记。在聊天摘要的上下文中，例如 Persona-Chat 数据集 [48]，提示和输出长度相对平衡，输入大小在 488–584 个令牌之间，平均输出为 44 个令牌。(3) 移动法学硕士现在支持长上下文窗口。例如，Qwen2-1.5B 等最新模型可以容纳最多 32K 个标记的上下文窗口，如表 1 所示。

2.2 机遇：移动 NPU

为了优化预填充延迟，`llm.npu` 利用了一个关键机会：现代移动 SoC 普遍包含移动

Table 1. The max context length for mobile-sized LLMs.

Model	Max Context	Year	Model	Max Context	Year
Opt-1.3B	2K	2022.5	TinyLLaMA-1.1B	2K	2023.9
StableLLM-3B	4K	2023.10	phi-2-2.7B	2K	2023.12
Gemma-2B	8K	2024.2	Qwen1.5-1.8B	32K	2024.2
Phi3-mini-3.8B	128K	2024.5	Qwen2-1.5B	32K	2024.6

**Figure 1.** Breakdown of end-to-end inference latency for UI automation, context-aware QA, and chat summaries. CPU evaluation uses llama.cpp as the on-device inference engine, while GPU evaluation uses TFLite as the simulation engine.**Table 2.** Specifications of well-known mobile NPUs provided by mainstream vendors.

Vendor	Latest NPU	SDK	Open	Group	INT8 Perf.
Qualcomm	Hexagon NPU [17]	QNN [25]	x	x	73 TOPS
Google	Edge TPU [19]	Edge TPU API [8]	x	x	4 TOPS
MediaTek	MediaTek APU 790 [12]	NeuroPilot [15]	x	N/A	60 TOPS
Huawei	Ascend NPU [7]	HiAI [10]	x	x	16 TOPS

"Open": Open-source?; "Group": Support per-group quantization MatMul?; "N/A": No available documents for public; "INT8 Perf": Int8 performance.

neural processing units (NPUs) that are well-suited for integer operations, such as INT8-based matrix multiplication. Table 2 summarizes the specifications of well-known mobile NPUs provided by mainstream vendors. For example, Qualcomm’s mobile SoCs feature Hexagon NPUs, achieving up to 73 trillion INT8 operations per second. According to AI-Benchmark [18], the Hexagon NPU in the Xiaomi 14 can infer the MobileNet-V2 model in just 0.6 ms, 23× faster than a mobile CPU and 3.2× faster than a mobile GPU.

Mobile NPU architecture. Mobile NPUs deliver significant performance benefits by single instruction multiple data (SIMD) architecture. For instance, Hexagon NPUs support 1024-bit INT8 vector arithmetic, allowing multiple SIMD instructions to execute in parallel. However, their floating-point computation capabilities are relatively weak compared to mobile GPUs. With clock frequencies between 500 and 750 MHz, mobile NPUs are more energy-efficient than mobile CPUs and GPUs. Additionally, unlike cloud GPUs that have separate physical memory, mobile NPUs are integrated within mobile SoCs, sharing the same physical memory with mobile CPUs, eliminating the need for memory copying during NPU execution.

Micro benchmark comparison. To evaluate the performance of INT8 MatMul on mobile NPUs, we conducted preliminary experiments on the Redmi K70 Pro using MatMul sizes commonly used in mobile LLMs, as summarized in

Table 3. Execution latency (ms) for various MatMul sizes on Redmi K70 Pro.

Matrix A	Matrix B	NPU INT8	CPU INT8	GPU FP16	NPU FP16
64*2048	2048*2048	0.9	4.2 (4.6×)	1.7 (1.9×)	252 (193×)
64*2048	2048*8192	1.5	6.8 (4.5×)	4.8 (3.2×)	986 (657×)
64*2048	2048*11008	2.0	11.6 (5.8×)	6.9 (3.5×)	1207 (603×)
32*4096	4096*4096	1.7	7.5 (4.4×)	3.1 (1.8×)	1054 (620×)
32*4096	4096*8192	2.9	13.1 (4.5×)	7.7 (2.6×)	2009 (692×)
32*4096	4096*11008	4.1	19.6 (4.8×)	10.4 (2.5×)	3112 (759×)

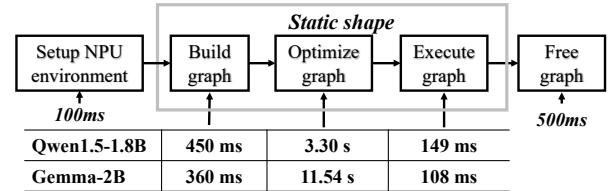
**Figure 2.** The workflow of executing DNNs on mobile NPUs, with latencies for each procedure on QNN [25].

Table 3. INT8 MatMul on mobile NPUs achieved a 4.5–5.8× speedup compared to CPU INT8 and a 1.8–3.5× improvement over GPU FP16. The performance gains increase with larger computational workloads. However, performing FP16 MatMul on the mobile NPU resulted in performance up to 159× slower than CPU INT8. These results align with the INT8 SIMD architecture of mobile NPUs, confirming that mobile NPUs are best suited for accelerating INT8 matrix multiplication.

DNN execution workflow on mobile NPUs. Executing DNNs on mobile NPUs involves configuring the NPU environment, creating the compute graph, optimizing the graph, executing the graph, and freeing the graph, as shown in Figure 2. Typically, creating and optimizing the compute graph are most time-consuming, where the former includes translating models into the NPU-required intermediate representation, and memory allocation, that takes 300–500ms, while the latter includes optimization for adjusting memory layout, execution order, and operator fusion, taking many seconds. In addition, the closed-source nature of NPU SDKs limits further adaptation for LLMs.

2.3 Gaps between LLMs and Mobile NPUs

Given its inherent advantages, we are surprised to find that none of existing DNN engines support LLM acceleration on mobile NPUs. We then dig into the underlying reasons and find a huge gap between existing mobile NPUs design and LLM inference pipeline.

• **LLM prefill phase relies on variable-length prompts, leading to excessive time spent on building and compiling the NPU graph.** As illustrated in Figure 2, before the compute graph can be executed on the mobile NPU, it must be built and optimized, a process taking tens of seconds. For instance, building the graph for the Gemma 2B

表 1. 移动规模 LLM 的最大上下文长度。

Model	Max Context	Year	Model	Max Context	Year
Opt-1.3B	2K	2022.5	TinyLLaMA-1.1B	2K	2023.9
StableLLM-3B	4K	2023.10	phi-2-2.7B	2K	2023.12
Gemma-2B	8K	2024.2	Qwen1.5-1.8B	32K	2024.2
Phi3-mini-3.8B	128K	2024.5	Qwen2-1.5B	32K	2024.6

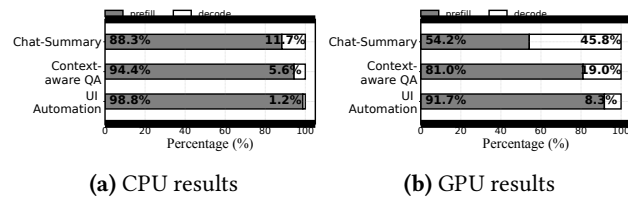


图 1. UI 自动化、上下文感知 QA 和聊天摘要的端到端推理延迟细分。CPU 评估使用 llama.cpp 作为设备端推理引擎，GPU 评估使用 TFLite 作为模拟引擎。

表2. 主流厂商提供的知名移动NPU规格。

Vendor	Latest NPU	SDK	Open	Group	INT8 Perf.
Qualcomm	Hexagon NPU [17]	QNN [25]	×	×	73 TOPS
Google	Edge TPU [19]	Edge TPU API [8]	×	×	4 TOPS
MediaTek	MediaTek APU 790 [12]	NeuroPilot [15]	×	N/A	60 TOPS
Huawei	Ascend NPU [7]	HiAI [10]	×	×	16 TOPS

“开放”：开源？；“Group”：支持每组量化 MatMul？ “N/A”：没有可供公众使用的文件；“INT8 Perf.”：Int8 性能。

神经处理单元（NPU）非常适合整数运算，例如基于 INT8 的矩阵乘法。表2总结了主流厂商提供的知名移动NPU的规格。例如，高通的移动 SoC 采用 Hexagon NPU，每秒可实现高达 73 万亿次 INT8 运算。根据 AI-Benchmark [18]，小米 14 中的 Hexagon NPU 只需 0.6 ms 即可推理出 MobileNet-V2 模型，比移动 CPU 快 23×，比移动 GPU 快 3.2×。

移动NPU架构。移动 NPU 通过单指令多数据 (SIMD) 架构提供显著的性能优势。例如，Hexagon NPU 支持 10 24 位 INT8 向量运算，允许多个 SIMD 指令并行执行。然而，与移动 GPU 相比，它们的浮点计算能力相对较弱。移动 NPU 的时钟频率在 500 至 750 MHz 之间，比移动 CPU 和 GPU 更节能。此外，与具有独立物理内存的云GPU不同，移动NPU集成在移动SoC中，与移动CPU共享相同的物理内存，从而无需在NPU执行期间进行内存复制。

微观基准比较。为了评估 INT8 MatMul 在移动 NPU 上的性能，我们使用移动 LLM 中常用的 MatMul 尺寸在 Redmi K70 Pro 上进行了初步实验，总结如下：

表 3. Redmi K70 Pro 上各种 MatMul 大小的执行延迟（毫秒）。

Matrix A	Matrix B	NPU INT8	CPU INT8	GPU FP16	NPU FP16
64*2048	2048*2048	0.9	4.2 (4.6×)	1.7 (1.9×)	252 (193×)
64*2048	2048*8192	1.5	6.8 (4.5×)	4.8 (3.2×)	986 (657×)
64*2048	2048*11008	2.0	11.6 (5.8×)	6.9 (3.5×)	1207 (603×)
32*4096	4096*4096	1.7	7.5 (4.4×)	3.1 (1.8×)	1054 (620×)
32*4096	4096*8192	2.9	13.1 (4.5×)	7.7 (2.6×)	2009 (692×)
32*4096	4096*11008	4.1	19.6 (4.8×)	10.4 (2.5×)	3112 (759×)

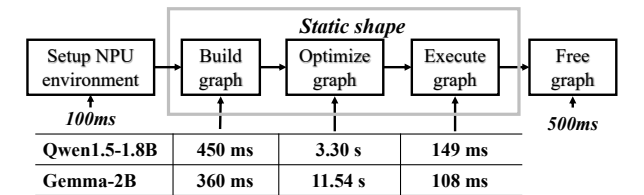


图 2. 在移动 NPU 上执行 DNN 的工作流程，以及 QNN 上每个过程的延迟 [25]。

表 3. 与 CPU INT8 相比，移动 NPU 上的 INT8 MatMul 实现了 4.5–5.8× 加速，比 GPU FP16 实现了 1.8–3.5× 改进。计算工作量越大，性能提升就越明显。然而，在移动 NPU 上执行 FP16 MatMul 导致性能比 CPU INT8 慢 1 59×。这些结果与移动 NPU 的 INT8 SIMD 架构一致，证实移动 NPU 最适合加速 INT8 矩阵乘法。

移动 NPU 上的 DNN 执行工作流程。在移动NPU上执行DNN涉及配置NPU环境、创建计算图、优化图、执行图和释放图，如图2所示。通常，创建和优化计算图是最耗时的，前者包括将模型转换为NPU所需的中间表示和内存分配，需要300-500ms，而后者包括调整内存布局、执行顺序和算子融合的优化，需要花费很多时间。秒。此外，NPU SDK 的闭源性质限制了 LLM 的进一步适配。

2.3 LLM 和移动 NPU 之间的差距

鉴于其固有的优势，我们惊讶地发现现有的 DNN 引擎都不支持移动 NPU 上的 LLM 加速。然后，我们深入研究根本原因，发现现有移动 NPU 设计与 LLM 推理流程之间存在巨大差距。

- LLM 预填充阶段依赖于可变长度的提示，导致在构建和编译 NPU 图上花费过多的时间。如图 2 所示，计算图在移动 NPU 上执行之前，必须对其进行构建和优化，该过程需要数十秒。例如，为 Gemma 2B 构建图表

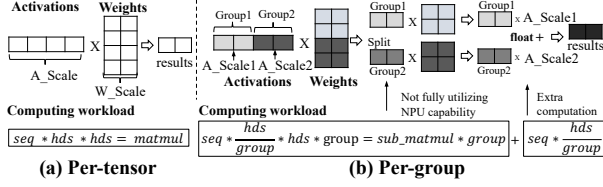


Figure 3. Per-tensor quantization MatMul and per-group quantization MatMul. *seq*, *hds*, *group* represent sequence length, hidden size, and group number, respectively.

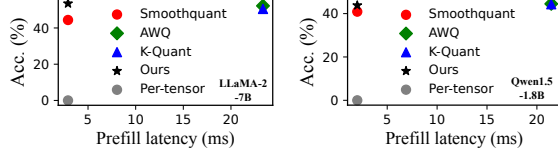


Figure 4. The prefill latency and accuracy on HelloSwag datasets among different quantization algorithms atop Xiaomi 14 using Qualcomm QNN framework.

model using QNN framework takes 360 ms, and graph optimization requires 11.54 seconds. Unlike CNN models, which are built and optimized once and can be executed multiple times with the same input shape, the LLM prefill phase must handle variable-length prompts, necessitating rebuilding and re-optimization of compute graphs for each inference. Consequently, using mobile NPUs in this scenario offers no performance benefit and is often slower than using a CPU.

• **The existence of activation outliers makes LLM difficult to quantize at whole-tensor level, yet a more fine-grained group-level quantization hampers NPU efficiency.** Our preliminary experiments, shown in Figure 4, indicate that two popular quantization algorithms (K-Quant [59] and AWQ [56]) incur significant inference overhead by $8.1\times$ – $10.7\times$, as compared to per-tensor quantization (Figure 3(a)). This is because algorithms like K-Quant and AWQ use fine-grained *per-group quantization* (Figure 3(b)) to maintain high accuracy. These algorithms divide activations and weights into multiple groups, each with an independent quantization scale. On NPUs, this approach requires dividing the MatMul operation into several sub-tensor MatMuls, which fails to fully leverage the capabilities of mobile NPUs. Additionally, it necessitates aggregating intermediate results with floating-point additions, resulting in extra floating-point computations. The exception is SmoothQuant [84], which uses per-tensor quantization but suffers from significant accuracy loss, such as a 3.9% and 8.4% drop on the HelloSwag dataset for the LLaMA-2-7B and Qwen1.5-1.8B model, respectively.

• **On-device LLM inference relies on floating-point operations, conflicting with the NPU’s design for INT8 acceleration.** Figure 5 illustrates a typical workflow for

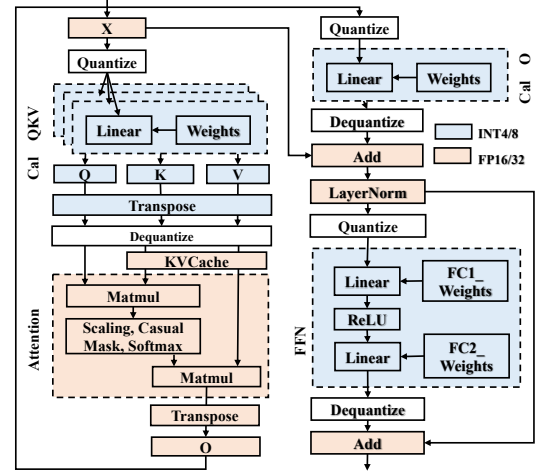


Figure 5. The workflow of quantized on-device LLM inference. Operations shown in blue are computed using INT4/INT8 formats, while those in orange are computed using float data formats.

Table 4. The operator data formats in the state-of-the-art quantization inference approaches. Each module in the first row is illustrated in Figure 5.

Quantization	Type	Acc.	Cal QKV	Atten.	Cal O	Norm.	FFN
K-Quant [59]	Per-Group	Low	INT8	FP16	INT8	FP16	INT8
GPTQ [35]	Per-Group	High	FP16	FP16	FP16	FP16	FP16
AWQ [56]	Per-Group	High	FP16	FP16	FP16	FP16	FP16
SmoothQuant [84]	Per-tensor	Low	INT8	FP16	INT8	FP16	INT8

"Atten.": Attention; "Norm.": Normalization.

quantized inference. To ensure inference accuracy, only linear layers (highlighted in blue) perform matrix multiplication in the INT8/INT4 data format. For these layers, the activation x is quantized to INT8/INT4 before performing the dot product with weights. Other operations, such as Attention and LayerNorm (highlighted in orange), are computed in floating-point format. Table 4 further summarize the operator data formats in state-of-the-art quantization inference algorithms. All of them depend on float Attention and normalization operations. Given that float operations dramatically degrade performance on mobile NPUs, these operations cannot be efficiently executed on the NPU with significant overhead.

3 11m.npu Design

3.1 Overview of 11m.npu

Design goal. 11m.npu aims to reduce prefill latency and energy consumption for mobile-sized LLMs through on-device NPU offloading. It supports various mobile-sized LLMs on devices and can be integrated as part of LLM-as-a-System-Service in mobile OS or mobile application services [99, 102]. **Workflow.** Figure 6 illustrates the workflow of 11m.npu. The key idea of 11m.npu is to maximize its execution on mobile NPU for integer operation acceleration; while keep

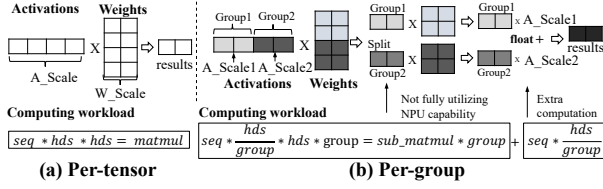


图 3. 每张量量化 MatMul 和每组量化 MatMul。seq、hds、group 分别表示序列长度、隐藏大小和组数。

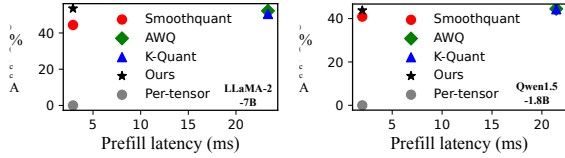


图 4. 使用 Qualcomm QNN 框架的小米 14 上不同量化算法之间 HelloSwag 数据集的预填充延迟和准确性。

使用 QNN 框架的模型需要 360 毫秒，图优化需要 11.54 秒。与 CNN 模型不同，CNN 模型构建和优化一次，并且可以使用相同的输入形状多次执行，LLM 预填充阶段必须处理可变长度的提示，从而需要为每个推理重建和重新优化计算图。因此，在这种情况下使用移动 NPU 不会带来任何性能优势，而且速度通常比使用 CPU 慢。

● 激活异常值的存在使得 LLM 难以在整个张量级别上进行量化，而更细粒度的组级别量化会阻碍 NPU 效率。我们的初步实验如图 4 所示，表明两种流行的量化算法（K-Quant [59] 和 AWQ [56]）与每张量量化相比（图 3(a)）会产生 $8.1\times$ – $10.7\times$ 的显著推理开销。这是因为 K-Quant 和 AWQ 等算法使用细粒度的每组量化（图 3(b)）来保持高精度。这些算法将激活和权重分为多个组，每个组都有独立的量化尺度。在 NPU 上，这种方法需要将 MatMul 操作划分为多个子张量 MatMul，这无法充分利用移动 NPU 的能力。此外，它还需要通过浮点加法来聚合中间结果，从而导致额外的浮点计算。SmoothQuant [84] 是个例外，它使用每张量量化，但精度损失显著，例如 LLaMA-2-7B 和 Qwen1.5-1.8B 模型的 HelloSwag 数据集分别下降了 3.9% 和 8.4%。

● 设备上 LLM 推理依赖于浮点运算，与 NPU 的 INT8 加速设计相冲突。图 5 说明了典型的工作流程

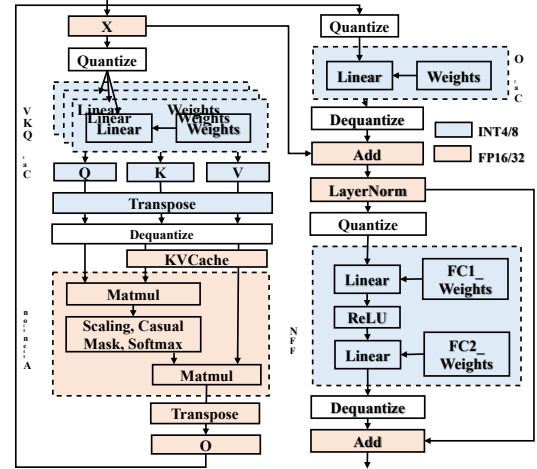


图 5. 量化设备上 LLM 推理的工作流程。蓝色显示的操作是使用 INT4/INT8 格式计算的，而橙色显示的操作是使用浮点数据格式计算的。

表 4. 最先进的量化推理方法中的运算符数据格式。第一行中的每个模块如图 5 所示。

Quantization	Type	Acc.	Cal QKV	Atten.	Cal O	Norm.	FFN
K-Quant [59]	Per-Group	Low	INT8	FP16	INT8	FP16	INT8
GPTQ [35]	Per-Group	High	FP16	FP16	FP16	FP16	FP16
AWQ [56]	Per-Group	High	FP16	FP16	FP16	FP16	FP16
SmoothQuant [84]	Per-tensor	Low	INT8	FP16	INT8	FP16	INT8

“注意”：注意；“规范”：规范化。

量化推理。为了确保推理准确性，只有线性层（以蓝色突出显示）以 INT8/INT4 数据格式执行矩阵乘法。对于这些层，在使用权重执行点积之前，激活 x 被量化为 INT8/INT4。其他操作，例如 Attention 和 LayerNorm（以橙色突出显示），以浮点格式计算。表 4 进一步总结了最先进的量化推理算法中的运算符数据格式。所有这些都依赖于浮动注意力和标准化操作。鉴于浮点运算会极大地降低移动 NPU 的性能，因此这些运算无法在 NPU 上高效执行，且开销较大。

3 llm.npu 设计

3.1 llm.npu 概述

设计目标。llm.npu 旨在通过设备上的 NPU 卸载来减少移动规模 LLM 的预填充延迟和能耗。它支持设备上的各种移动大小的 LLM，并且可以作为 LLM 即系统服务的一部分集成到移动操作系统或移动应用程序服务中 [99, 102]。工作流程。图 6 说明了 llm.npu 的工作流程。llm.npu 的核心思想是最大化其在移动 NPU 上的执行，以实现整数运算加速；同时保持

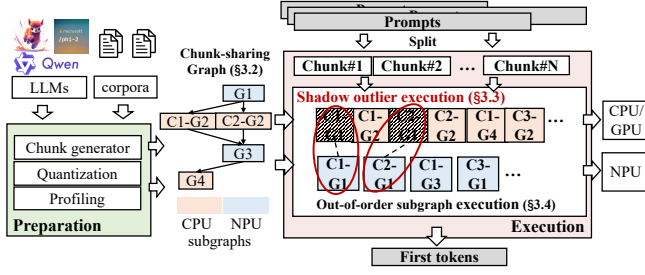


Figure 6. The workflow of 11m.npu.

necessary floating point operations on CPU/GPU to not compromise accuracy. To enable more efficient NPU offloading, 11m.npu re-constructs the prompt and model in following ways: (1) At *prompt level*: variable-length prompt is reduced into multiple fixed-sized chunks with data dependency preserved; (2) At *block level*: Transformer block is scheduled into CPU/GPU and NPU based on their hardware infinity and accuracy sensitivity; (3) At *tensor level*: important outliers are identified and extracted to run on CPU/GPU.

- **Preparation stage.** 11m.npu first uses an enhanced per-tensor quantization algorithm to quantize LLMs into W8A8 format. The quantization algorithm differs from existing ones as it filters out most unimportant activation outliers and extracts the rest of them into independent, lightweight operators that are complementary to the original one. 11m.npu also generates fixed-length *chunk-sharing graphs* (§3.2) to efficiently handle variable-length prompts.

- **Execution stage.** When receiving a prompt, 11m.npu divides it into fixed-sized chunks and processes them causally. These chunk graphs will be split into subgraphs scheduled onto the CPU/GPU and NPU according to their data formats for efficient execution. To preserve accuracy, certain INT8-based linear layers undergo sparse float outlier shadow execution on the CPU/GPU in parallel to compensate for quantization errors from outliers (§3.3). To enhance execution efficiency, 11m.npu judiciously schedules the chunks in out-of-order manner (§3.4).

3.2 Chunk-sharing graph execution

To tackle the dynamic prompt length challenge, an intuitive solution is to set a fixed length compute graph ahead and use padding [3, 67, 77]. However, this method lacks flexibility and excessive padding wastes compute resources.

Chunk-wise prefill. To enhance flexibility and minimize padding for variable-length prompts, we recognize that processing a long prompt in a LLM is equivalent to processing several split sub-prompts, or “chunks”, causally. This is feasible because popular LLMs use a decoder-only architecture, where the result of the i -th token depends only on the preceding tokens. To that end, 11m.npu first pre-builds and pre-optimizes fixed-length chunk-based NPU compute graph at the preparation stage. During inference, 11m.npu splits long

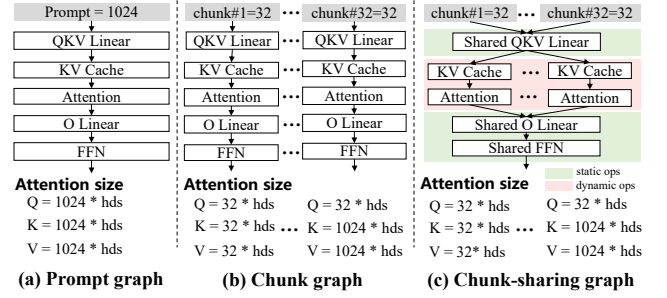


Figure 7. The illustration of prompt graph, chunk graph and chunk-sharing graph. The chunk length is 32.

prompts into several chunks and processes them using these pre-built chunk graphs, as illustrated in Figure 7(b).

However, solely using chunk graphs is not scalable, as 11m.npu would need to store numerous distinct chunk graphs in memory, significantly increasing memory overhead. This is because different chunk graphs have attention operators of varying sizes. For instance, considering a prompt length of 1024 and a chunk length of 32, the QKV dimension sizes of the attention operators for the first chunk are all $32 * hds$, while for the last chunk they are $32 * hds$, $1024 * hds$, and $1024 * hds$ respectively, as shown in Figure 7(b).

Chunk-sharing graph. 11m.npu introduces a *chunk-sharing graph*, shown in Figure 7(c), based on the insight that LLM operators fall into two distinct categories: (1) static operators (in green), such as Linear and LayerNorm, which depend only on the chunk length and can be shared across different chunks; and (2) dynamic operators (in red), such as Attention, which depend on both chunk length and chunk sequence and cannot be shared among different chunks. Consequently, 11m.npu divides the LLM into several subgraphs based on the shareability of operators. The shared subgraphs are built and optimized once, whereas non-shared subgraphs are constructed individually for different chunks. During the prefill phase, activations from different chunks pass through the same static operator subgraphs while dynamically selecting the appropriate dimension-specific dynamic operators. This method significantly reduces memory overhead and enhances scalability, as most dynamic operators, like Attention, do not contain weights, requiring only activation buffers.

Our experiments show that 120 out of 144 subgraphs can be shared in Qwen1.5-1.8B models, reducing memory consumption by up to 75% (7.2GB) for a prompt length as 1024 and a chunk length as 256.

We also conducted extensive experiments on selecting a proper chunk length. The results of two popular LLMs (Qwen1.5-1.8B and Gemma-2B) on Xiaomi 14 device is illustrated in Figure 8. Based on the observations, 11m.npu empirically chooses a chunk length of 256 for Xiaomi 14 device, which effectively utilizes the capabilities of mobile

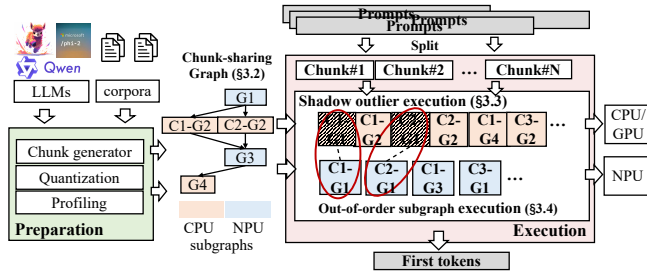


图 6. llm.npu 的工作流程。

CPU/GPU 上必要的浮点运算，以免影响精度。为了实现更高效的 NPU 卸载，llm.npu 通过以下方式重新构建提示和模型：（1）在提示级别：可变长度提示被缩减为多个固定大小的块，并保留数据依赖性；（2）块级：Transformer 块根据硬件无限性和精度敏感度被调度到 CPU/GPU 和 NPU 中；（3）在张量级别：识别并提取重要的异常值以在 CPU/GPU 上运行。

- 准备阶段。llm.npu 首先使用增强的每张量量化算法将 LLM 量化为 W8A8 格式。量化算法与现有算法不同，因为它过滤掉最不重要的激活异常值，并将其余的提取到独立的、轻量级的算子中，作为原始算子的补充。llm.npu 还生成固定长度的块共享图 (§3.2)，以有效地处理可变长度的提示。

- 执行阶段。当收到提示时，llm.npu 将其分为固定大小的块并按因果关系进行处理。这些块图将根据数据格式分成调度到 CPU/GPU 和 NPU 上的子图，以实现高效执行。为了保持准确性，某些基于 INT8 的线性层在 CPU/GPU 上并行进行稀疏浮点离群值阴影执行，以补偿离群值的量化误差 (§3.3)。为了提高执行效率，llm.npu 明智地以无序方式调度块 (§3.4)。

3.2 块共享图执行

为了解决动态提示长度的挑战，一个直观的解决方案是预先设置一个固定长度的计算图并使用填充[3,67,77]。然而，这种方法缺乏灵活性，并且过多的填充会浪费计算资源。

分块预填充。为了增强灵活性并最大限度地减少可变长度提示的填充，我们认识到在 LLM 中处理长提示相当于按因果关系处理多个分割的子提示或“块”。这是可行的，因为流行的 LLM 使用仅解码器架构，其中第 i 个令牌的结果仅取决于前面的令牌。为此，llm.npu 首先在准备阶段预构建和预优化基于固定长度块的 NPU 计算图。在推理过程中，llm.npu 进行长分割

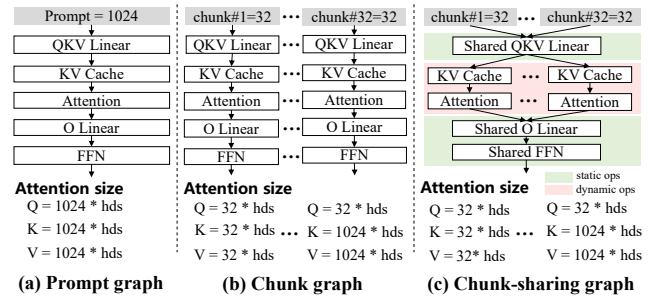


图7.提示图、块图和块共享图的图示。块长度为 32。

提示分成几个块并使用这些预先构建的块图处理它们，如图 7(b) 所示。

然而，单独使用块图是不可扩展的，因为 llm.npu 需要在内存中存储大量不同的块图，从而显著增加内存开销。这是因为不同的块图具有不同大小的注意力运算符。例如，考虑到提示长度为 1024，块长度为 32，第一个块的注意力算子的 QKV 维度大小都是 $32 * hds$ ，而最后一个块的 QKV 维度大小分别是 $32 * hds$ 、 $1024 * hds$ 和 $1024 * hds$ ，如图 7(b) 所示。

块共享图。llm.npu 引入了一个块共享图，如图 7 (c) 所示，基于 LLM 运算符分为两个不同类别的见解：

- （1）静态运算符（绿色），例如 Linear 和 LayerNorm，它们仅依赖于块长度并且可以在不同块之间共享；
- （2）动态算子（红色），例如 Attention，它依赖于块长度和块序列，并且不能在不同块之间共享。因此，llm.npu 根据操作员的可共享性将 LLM 分为几个子图。共享子图构建并优化一次，而非共享子图针对不同块单独构建。在预填充阶段，来自不同块的激活通过相同的静态运算符子图，同时动态选择适当的特定于维度的动态运算符。这种方法显著减少了内存开销并增强了可扩展性，因为大多数动态运算符（例如注意力）不包含权重，只需要激活缓冲区。

我们的实验表明，144 个子图中的 120 个可以在 Qwen1.5-1.8B 模型中共享，对于提示长度为 1024 和块长度为 256 的情况，内存消耗最多可减少 75% (7.2GB)。

我们还进行了广泛的实验来选择合适的块长度。两种流行的 LLM (Qwen1.5-1.8B 和 Gemma-2B) 在小米 14 设备上的结果如图 8 所示。根据观察，llm.npu 经验性地为小米 14 设备选择了 256 的块长度，这有效地利用了移动设备的能力

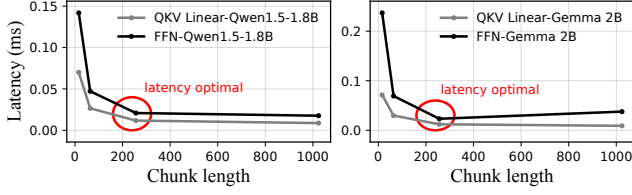


Figure 8. The per-token latency of QKV Linears and FFN under different chunk lengths.

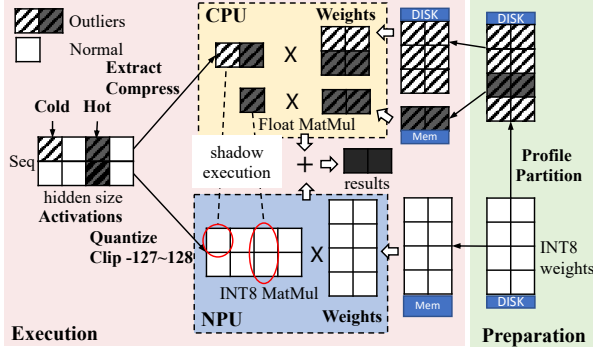


Figure 9. The workflow of shadow outlier execution.

NPUs while reducing intra-chunk padding. In practice, such profiling needs to be performed across different NPUs.

3.3 Shadow outlier execution

To enable NPU-friendly, per-tensor activation quantization without compromising LLM accuracy, `llm.npu` adopts a novel approach termed *shadow outlier execution*. As shown in Figure 9, `llm.npu` extracts the activation channels with outliers at runtime into a more compact tensor, executes it on CPU, and merges it back to the outcome of original operator on NPU. This procedure can be formulated as follows:

$$\begin{aligned} \frac{x}{s} \odot w &= \left\{ \min \left[\max \left(\frac{x}{s}, -127 \right), 128 \right] + \left\lfloor \frac{x}{s} / 128 \right\rfloor \times 128 \right\} \odot w \\ &= \min \left[\max \left(\frac{x}{s}, -127 \right), 128 \right] \odot w \quad \text{on NPU} \\ &+ \text{extract} \left(\left\lfloor \frac{x}{s} / 128 \right\rfloor \times 128 \right) \odot w \quad \text{on CPU} \end{aligned} \quad (1)$$

where x , w , s , $\odot$, and *extract* represent the original float activation, INT8 weights, the quantization scale factor, the MatMul operation, and the function of extracting activation outliers into a more compact tensor, respectively. Specifically, the MatMul $\frac{x}{s} \odot w$ can be equivalently divided into the sum of two parts according to the associative law: (1) *Mobile NPU for MatMul within the scale*. `llm.npu` first quantizes and rounds x to the range of -127 to 128 based on the scale factor s . It then obtains intermediate results by performing a standard W8A8 per-tensor MatMul with weights w . (2) *Mobile CPUs/GPUs for MatMul beyond the scale*. `llm.npu` calculates the partial values exceeding s . Since these outliers are rare, `llm.npu`

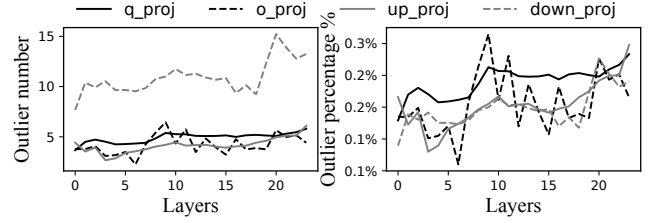


Figure 10. The average number and percentage of outlier channels per layer on Qwen1.5-1.8B model using the wikitext dataset under 2048 inference. **Less than 0.3% channels have outliers during one inference.**

extracts these values from the tensor, compresses them into a dense tensor, and performs a MatMul with weights w .

Since outliers are very sparse (around 5–15 channels, accounting for only 0.1%–0.3% of total channels, as shown in Figure 10), the shadow execution on CPU is much faster than the execution of original tensor on NPU, and its execution time can be totally hidden by overlapping. To further minimize the overhead of this extra process, `llm.npu` determines an outlier threshold (i.e., s in Equation 1) by profiling a large corpora at offline, thereby can identify the outliers by simply comparing the activation numbers to this threshold. The design of shadow outlier execution is compatible with any per-tensor quantization algorithms, and the current prototype of `llm.npu` is based on the simple max-min symmetry quantization [47].

While the shadow outlier execution seems to have well balanced the NPU affinity and LLM accuracy, two more crucial issues need to be addressed to make it practical. First, though mobile SoC uses a unified memory chip for heterogeneous processors, they use separated memory space. To enable shadow execution of activation outliers, `llm.npu` has to keep another copy of each MatMul weights on CPU’s memory space. This increases the memory footprint by almost 2 times. Second, while the execution of outlier is fast even on CPU, the synchronization of the reduced sum between CPU and NPU still takes non-trivial overhead, e.g., 29.7% end-to-end latency and 20.1% energy consumption on Qwen1.5-1.8B.

Most outliers tend to appear in a small set of channel positions. We observe that, while outliers appear in a wide range of channel positions during processing a long prompt (e.g., 78%), such appearance is highly skewed – few channels dominate the appearance of outliers. For instance, as shown in Figure 11, less than 3% of channels contribute to the majority of outliers (over 80%) across various inputs on Qwen1.5-1.8B and LLaMA-2-7B models. Therefore, for shadow outlier execution, `llm.npu` only keeps the tensor weights that need to be used by those “hot channels” in CPU memory space, and retrieve the rest of them from disk if outliers on those positions are extracted (which is rare) at

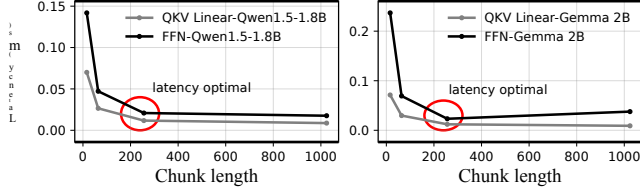


图 8. QKV Linears 和 FFN 在不同块长度下的每个令牌延迟。

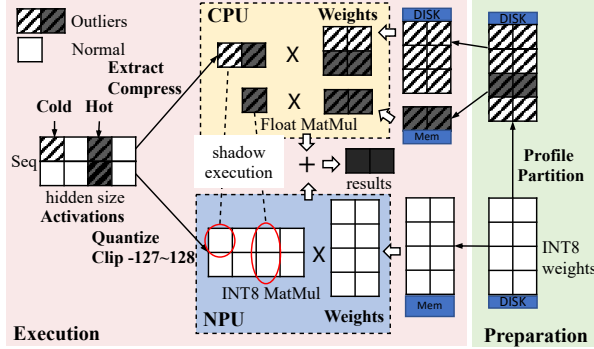


图 9. 影子异常值执行的工作流程。

NPU 同时减少块内填充。实际上，这种分析需要在不同的 NPU 上执行。

3.3 影子异常值执行

为了在不影响 LLM 精度的情况下实现 NPU 友好的每张量激活量化，llm.npu 采用了一种称为阴影异常值执行的新颖方法。如图 9 所示，llm.npu 在运行时将具有异常值的激活通道提取为更紧凑的张量，在 CPU 上执行，然后将其合并回 NPU 上原始算子的结果。该程序可以表述如下：

$$\begin{aligned} \frac{x}{s} \odot w &= \left\{ \min \left[\max \left(\frac{x}{s}, -127 \right), 128 \right] + \left\lfloor \frac{x}{s} / 128 \right\rfloor \times 128 \right\} \odot w \\ &= \min \left[\max \left(\frac{x}{s}, -127 \right), 128 \right] \odot w \quad \text{on NPU} \\ &\quad + \text{extract} \left(\left\lfloor \frac{x}{s} / 128 \right\rfloor \times 128 \right) \odot w \quad \text{on CPU} \end{aligned} \quad (1)$$

其中 x 、 w 、 s 、 $\odot$ 和 extract 分别表示原始浮点激活、INT8 权重、量化比例因子、MatMul 操作以及将激活异常值提取到更紧凑的张量中的函数。具体来说，根据结合律， $\text{MatMul}_s^x \odot w$ 可以等价地分为两部分之和：(1) 尺度内 MatMul 的移动 NPU。llm.npu 首先根据比例因子 s 将 x 量化并舍入到 -127 到 128 的范围。然后，它通过使用权重 w 执行标准 W8A8 每个张量 MatMul 来获得中间结果。(2) 超出规模的用于 MatMul 的移动 CPU/GPU。llm.npu 计算超过 s 的部分值。由于这些异常值很少见，llm.npu

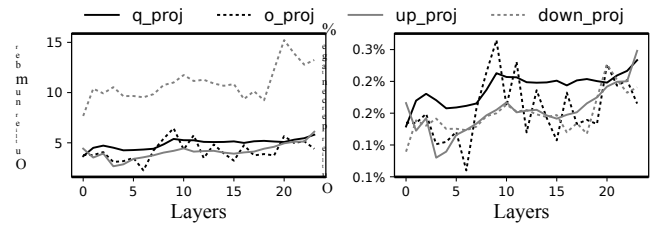


图 10. 使用 wikitext 数据集在 2048 次推理下的 Qwen1.5-1.8B 模型上每层离群值通道的平均数量和百分比。在一次推理过程中，只有不到 0.3% 的通道存在异常值。

从张量中提取这些值，将它们压缩为稠密张量，并使用权重 w 执行 MatMul。

由于异常值非常稀疏（大约 5-15 个通道，仅占总通道的 0.1%-0.3%，如图 10 所示），CPU 上的影子执行速度比 NPU 上原始张量的执行速度快得多，并且其执行时间可以通过重叠完全隐藏。为了进一步最小化这个额外过程的开销，llm.npu 通过离线分析大型语料库来确定异常值阈值（即等式 1 中的 s ），从而可以通过简单地将激活数与该阈值进行比较来识别异常值。阴影异常值执行的设计与任何每张量量化算法兼容，并且 llm.npu 的当前原型基于简单的最大最小对称量化[47]。

虽然影子异常值执行似乎很好地平衡了 NPU 亲和力和 LLM 准确性，但还需要解决两个更关键的问题才能使其实用。首先，虽然移动 SoC 对异构处理器使用统一的存储芯片，但它们使用独立的存储空间。为了启用激活异常值的影子执行，llm.npu 必须在 CPU 内存空间上保留每个 MatMul 权重的另一个副本。这使内存占用增加了近 2 倍。其次，虽然异常值的执行即使在 CPU 上也很快，但 CPU 和 NPU 之间的减少总和的同步仍然需要不小的开销，例如，Qwen1.5-1.8B 上的端到端延迟为 29.7%，能耗为 20.1%。

大多数异常值往往出现在一小部分通道位置中。我们观察到，虽然在处理长提示期间（例如 78%），异常值出现在各种通道位置，但这种外观是高度倾斜的——很少有通道主导异常值的出现。例如，如图 11 所示，在 Qwen1.5-1.8B 和 LLaMA-2-7B 模型的各种输入中，不到 3% 的通道对大多数异常值（超过 80%）有贡献。因此，对于影子异常值执行，llm.npu 仅将那些“热通道”需要使用的张量权重保留在 CPU 内存空间中，如果在这些位置上提取异常值（这种情况很少见），则从磁盘检索其余部分。

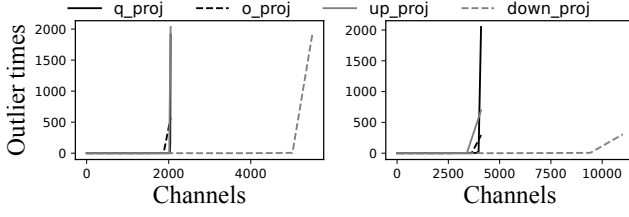


Figure 11. The outlier times per channel on Qwen1.5-1.8B model using the wikitext dataset under 2048 inference. **Less than 3% channels contribute over 80% outliers.**

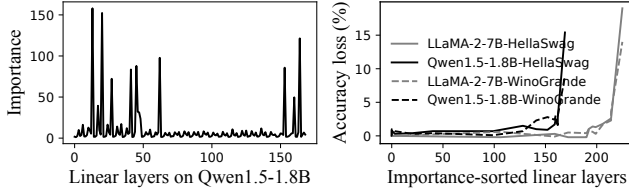


Figure 12. Left: Outlier importance of different layers in Qwen1.5-1.8B model. Right: Relationship between accuracy and pruned layers on HellaSwag and Winograde datasets.

runtime. Note that the weights retrieval can be also overlapped with the NPU execution of the original MatMul. This approach reduces the memory overhead of shadow execution by 34.3% with negligible latency.

Most outliers can be pruned without impacts on accuracy. Surprisingly, we observe that the activation outliers on most MatMul operators are unimportant to the LLM accuracy and can be simply removed. Here, the importance of an outlier is measured as the ratio between the largest outlier and the quantization scale (s in Equation 1). The larger ratio indicates a more dispersed activation distribution, resulting in more significant quantization errors. 11m.npu profiles these outlier importance using large corpora data at offline stage, (Figure 12), and prunes most of unimportant layers' outliers. Typically, we observed layers near the inputs and outputs have a higher importance. This is because layers close to inputs are easily influenced by the tokens disparity, exhibiting greater fluctuations, while layers approaching to outputs easily accumulate the errors from the shallow layers. Based on the observation, 11m.npu prunes the outliers of top 85% most unimportant layers through offline profiling, so that the CPU-NPU synchronization is eliminated.

3.4 Out-of-order subgraph execution

As elaborated in (§2.3), LLM quantization algorithms cannot fully eliminate floating point operations, 11m.npu thereby divides its execution flow into NPU and CPU/GPU collaboratively. Typically, LayerNorm, Attention, as well as the shadow outlier computation are placed on the CPU/GPU;

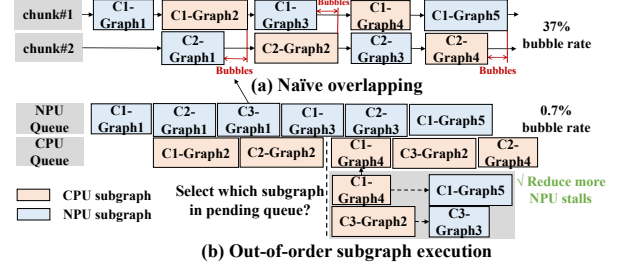


Figure 13. Out-of-order subgraph execution.

while the other linear layers are processed on the NPU. However, we found simply overlapping their execution is inefficient, resulting in large execution bubbles (37% bubble rate in critical path), as illustrated in Figure 13(a).

Out-of-order execution. To reduce these execution bubbles, 11m.npu is guided by a key insight that, after being partitioned at both chunk and subgraphs levels, the LLM subgraphs can be scheduled in an out-of-order manner. More specifically, any input-ready subgraph can be executed without strictly following the chunk sequence. For instance, the first subgraph of the third chunk (C3-Graph1) can be executed during the bubble period when C2-Graph1 finishes.

To preserve correctness, 11m.npu considers two types of dependency: (1) *Cross-chunk dependency*. Operators like Attention rely on data from previous chunks. This means the i -th chunk j -th subgraph $G_{i,j}$ depends on the $j-1$ -th subgraph of the $0, 1, \dots, i-1$ chunks:

$$G_{i,j} \leftarrow G_{0,j-1}, G_{1,j-1}, \dots, G_{i-1,j-1} \quad (2)$$

(2) *Intra-chunk dependency*. Operators like LayerNorm, Linear, and Quantize rely only on previous subgraphs within the same chunk. Therefore, the i -th chunk's j -th subgraph $G_{i,j}$ depends on the $j-1$ -th subgraph of the same chunk:

$$G_{i,j} \leftarrow G_{i,j-1} \quad (3)$$

As mobile processors are weak at parallelism and preemption [40, 86, 97], to ensure efficiency, a processor is capable of executing only one subgraph at any given time.:

$$\sum_{i=0}^N \sum_{j=0}^M P_{i,j,t} = 1, \forall t \quad (4)$$

where $P_{i,j,t} = 1$ indicates that subgraph $G_{i,j}$ is running on processor P at time t , and N and M represent the maximum number of chunks and subgraphs, respectively. 11m.npu aims to find an execution order minimizing the total execution time of all subgraphs under these constraints. Unfortunately, this scheduling problem can be reduced to a classical NP-Hard Traveling Salesman Problem [44]. Moreover, because the number of chunks varies with user prompts, an optimal scheduling strategy cannot be generated offline.

Instead, 11m.npu utilizes an online heuristic algorithm. The key idea is to focus not on the execution time of the

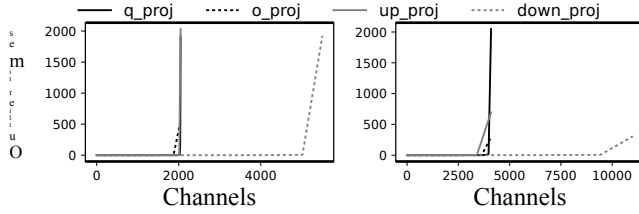


图 11. 使用 wikitext 数据集在 2048 推断下的 Qwen1.5-1.8B 模型上每个通道的异常值时间。不到 3% 的渠道贡献了超过 80% 的异常值。

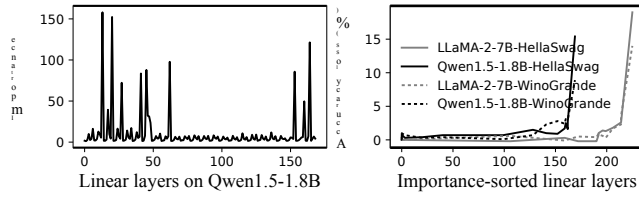


图 12. 左: Qwen1.5-1.8B 模型中不同层的异常重要性。右图: HellaSwag 和 Winograde 数据集上的准确度与修剪层之间的关系。

运行时。请注意，权重检索也可以与原始 MatMul 的 NPU 执行重叠。这种方法将影子执行的内存开销减少了 34.3%，并且延迟可以忽略不计。

大多数异常值都可以被修剪，而不会影响准确性。令人惊讶的是，我们观察到大多数 MatMul 算子上的激活异常值对 LLM 精度并不重要，可以简单地删除。这里，离群值的重要性通过最大离群值与量化尺度（等式 1 中的 s ）之间的比率来衡量。比率越大表示激活分布越分散，从而导致量化误差越显著。llm.npu 在离线阶段使用大型语料库数据来描述这些异常值的重要性（图 12），并修剪大部分不重要层的异常值。通常，我们观察到输入和输出附近的层具有更高的重要性。这是因为靠近输入的层很容易受到令牌差异的影响，表现出更大的波动，而靠近输出的层很容易积累来自浅层的误差。根据观察，llm.npu 通过离线分析修剪了前 85% 最不important 层的异常值，从而消除了 CPU-NPU 同步。

3.4 子图乱序执行

正如 (§2.3) 中所述，LLM 量化算法无法完全消除浮点运算，因此 llm.npu 将其执行流程划分为 NPU 和 CPU/GPU 协作。通常，LayerNorm、Attention 以及阴影异常值计算都放在 CPU/GPU 上；

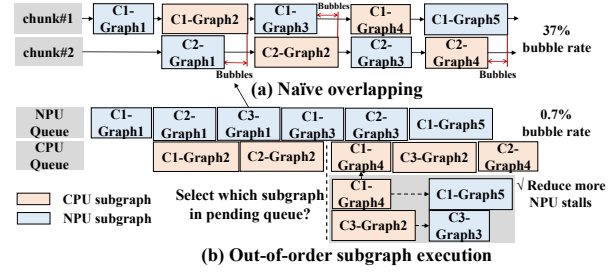


图 13. 无序子图执行。

而其他线性层则在 NPU 上处理。然而，我们发现简单地重叠它们的执行效率低下，导致较大的执行泡沫（关键路径中的泡沫率为 37%），如图 13 (a) 所示。

无序执行。为了减少这些执行气泡，llm.npu 以一个关键见解为指导，即在块和子图级别上进行分区后，LLM 子图可以以无序方式进行调度。更具体地说，任何输入就绪的子图都可以在不严格遵循块序列的情况下执行。例如，当 C2-Graph1 完成时，可以在冒泡期间执行第三个块（C3-Graph1）的第一个子图。

为了保持正确性，llm.npu 考虑两种类型的依赖关系：

- (1) 跨块依赖关系。像注意力这样的运算符依赖于先前块的数据。这意味着第 i 个块 j 个子图 $G_{i,j}$ 取决于 $0, 1, \dots, i-1$ 个块的第 $j-1$ 个子图：

$$G_{i,j} \leftarrow G_{0,j-1}, G_{1,j-1}, \dots, G_{i-1,j-1} \quad (2)$$

- (2) 块内依赖。LayerNorm、Linear 和 Quantize 等运算符仅依赖于同一块内的先前子图。因此，第 i 个块的第 j 个子图 $G_{i,j}$ 取决于同一块的第 $j-1$ 个子图：

$$G_{i,j} \leftarrow G_{i,j-1} \quad (3)$$

由于移动处理器在并行性和抢占方面较弱[40,86,97]，为了确保效率，处理器在任何给定时间只能执行一个子图：

$$\sum_{i=0}^N \sum_{j=0}^M P_{i,j,t} = 1, \forall t \quad (4)$$

其中 $P_{i,j,t} = 1$ 表示子图 $G_{i,j}$ 在时间 t 在处理器 P 上运行，而 N 和 M 分别表示块和子图的最大数量。llm.npu 的目标是找到一个执行顺序，在这些约束下最小化所有子图的总执行时间。不幸的是，这个调度问题可以简化为经典的 NP 困难旅行商问题 [44]。而且，由于块的数量随用户提示而变化，无法离线生成最优调度策略。

相反，llm.npu 使用在线启发式算法。关键思想是不要关注执行时间

subgraph g , but on how executing g contributes to reducing NPU stalls, motivated by the observation that during the prefill phase, NPU execution time often dominates inference latency, being the critical path. For instance, with a prompt length of 256 using the Qwen1.5-1.8B model, NPU execution takes 315ms, about twice that of the CPU.

Specifically, we define a subgraph g 's contribution to reduce NPU stalls as follows: If subgraph g is to be executed on the CPU/GPU, let S be the set of new subgraphs that can be executed after g is completed. S will be executed on the NPU. A longer execution time of S is beneficial for reducing NPU stalls. Thus, g 's contribution is defined as the total execution time of S . Conversely, if g is executed on the NPU, a shorter execution time of S is beneficial, with the negative value of S 's execution time as g 's contribution, formulated as:

$$C = \begin{cases} \sum T_i, \forall i \in S \text{ if } g \text{ is on the CPU/GPU} \\ -\sum T_i, \forall i \in S \text{ if } g \text{ is on the NPU} \end{cases} \quad (5)$$

where T is the subgraph execution time. `11m.npu` always chooses the subgraph with the largest C , meaning the subgraph g with S having the longest execution time on the NPU or the shortest execution time on the CPU/GPU.

In a nut shell, `11m.npu` profiles all the subgraph execution time and their dependency offline at the preparation stage. During the prefill stage, it calculates all the pending subgraphs C value and selects one with maximum C to run, with microsecond-level performance overhead.

4 Implementation and Evaluation

We have fully implemented `11m.npu` for Qualcomm Hexagon NPUs, comprising 10K lines of code in C/C++ and assembly language. We choose Qualcomm SoCs as the target platform for its popularity on mobile devices and powerful NPU capacity. Qualcomm Hexagon is also the only mobile NPU with an open instruction set architecture. `11m.npu` is built on the MLLM [23], one state-of-the-art mobile LLM engines, and QNN framework [25], the Qualcomm Neural Processing SDK. It supports standard LLM formats exported from Hugging Face [21]. To facilitate LLM execution, we implemented specific operators like KVCache, SiLU, RMSNorm, ROPE, and etc, in addition to what have been supported by QNN. To reduce context switching overhead between CPUs/GPUs and NPUs, `11m.npu` leverages shared buffers to synchronize intermediate results from different processors. For end-to-end inference, `11m.npu` is compatible with any decoding engine and utilizes the MLLM CPU backend for decoding stage as easy implementation, with a default chunk length of 256. The default pruning rate for outlier layers is 85%. `11m.npu` does not consider resource detection and contention in the current implementation.

The prototype further incorporates two optimizations. (1) Our extensive experiments show that mobile NPUs often favor tensor sizes (e.g., equal “height” and “width”) in CNN architectures. For example, a linear layer with weights of

2048×2048 produces the same results for inputs of $1024 \times 1 \times 2048$ and $32 \times 32 \times 2048$, but using $32 \times 32 \times 2048$ reduces execution latency by 1.62×. Therefore, `11m.npu` profiles all possible equivalent shapes for linear layers during the preparation stage and selects the most efficient one. (2) Mobile NPUs typically access limited memory regions (e.g., 4GB for Hexagon NPU), which can be smaller than the size of LLM weights. To maximize prefill acceleration within this limited memory, `11m.npu` prioritizes executing computationally intensive tasks, such as FFN, on the NPU to enhance efficiency.

4.1 Experiment setups

Hardware setup. We test `11m.npu` on two smartphones with different Qualcomm SoCs: Redmi K70 Pro (Snapdragon 8gen3, 24GB memory) and Redmi K60 Pro (Snapdragon 8gen2, 16GB memory). All devices run Android OS 13.

Models and datasets. We conducted tests using a variety of typical mobile-sized LLMs: Qwen1.5-1.8B [27], Gemma-2B [9], Phi2-2.7B [16], LLaMA2-Chat-7B [11], and Mistral-7B [14]. To evaluate `11m.npu`'s quantization accuracy, we employed widely recognized LLM benchmarks, including LAMBADA [68], HellaSwag [103], WinoGrande [4], OpenBookQA [62] and MMLU [43]. For inference speed experiments, we selected retrieval-based datasets from Longbench, 2wikimqa and TriviaQA [28], for simulating context-aware generate tasks like automated email reply. Additionally, we assessed `11m.npu` in screen question-answering and mapping instruction to UI action scenarios using DroidTask datasets [22] to simulate agent-based UI automation tasks.

Baselines. We mainly compare `11m.npu` with 5 baselines, including 3 widely used mobile LLM engines (TFLite [57], MNN [49], and llama.cpp [59]). Those engines support only mobile CPU and GPU; 2 advanced baselines are MLC-LLM [75], an LLM compiler for on-device GPUs, and PowerInfer-v2, which also utilizes mobile NPUs to accelerate prefilling [94]. Since PowerInfer-v2 is not open-sourced, we use the reported data from its paper. To be noted, those baselines often support only a subset of 5 LLMs we evaluated.

Metrics. We mainly measure LLM inference accuracy, prefill latency, prefill energy consumption, prefill memory consumption and end-to-end inference latency. The energy consumption is obtained through `/sys/class/power_supply` in Android OS by profiling every 100ms. All experiments are repeated three times and we report the average numbers.

4.2 Prefill performance.

We evaluate the `11m.npu`'s prefill performance (speed and energy consumption) at prompt lengths of 64, 256 and 1024 tokens on two devices, as shown in Figure 14 and 15. Despite outlier variations across datasets, the overall impact on prefill performance is minimal, so we report results from the LongBench dataset. The results show that `11m.npu` *consistently outperforms all baselines across both metrics, with benefits becoming more pronounced as prompt length increases.*

子图 g ，但关于执行 g 如何有助于减少 NPU 停顿，其动机是观察到在预填充阶段，NPU 执行时间通常主导推理延迟，是关键路径。例如，使用 Qwen1.5-1.8B 模型，提示长度为 256，NPU 执行需要 315ms，大约是 CPU 的两倍。

具体来说，我们定义子图 g 对减少 NPU 停顿的贡献如下：如果子图 g 要在 CPU/GPU 上执行，则令 S 为在 g 完成后可以执行的新子图的集合。 S 将在 NPU 上执行。 S 较长的执行时间有利于减少 NPU 停顿。因此， g 的贡献被定义为 S 的总执行时间。相反，如果 g 在 NPU 上执行，则 S 的执行时间较短是有利的， S 的执行时间的负值作为 g 的贡献，公式为：

$$C = \begin{cases} \sum T_i, \forall i \in S \text{ if } g \text{ is on the CPU/GPU} \\ -\sum T_i, \forall i \in S \text{ if } g \text{ is on the NPU} \end{cases} \quad (5)$$

其中 T 是子图执行时间。llm.npu 始终选择具有最大 C 的子图，这意味着具有 S 的子图 g 在 NPU 上的执行时间最长，或者在 CPU/GPU 上的执行时间最短。

简而言之，llm.npu 在准备阶段离线分析所有子图执行时间及其依赖关系。在预填充阶段，它会计算所有待处理的子图 C 值，并选择具有最大 C 的子图来运行，性能开销为微秒级。

4 实施与评估

我们已经为 Qualcomm Hexagon NPU 全面实现了 llm.npu，其中包含 10K 行 C/C++ 和汇编语言代码。我们选择高通 SoC 作为目标平台，因为它在移动设备上的受欢迎程度和强大的 NPU 能力。Qualcomm Hexagon 也是唯一一款具有开放指令集架构的移动 NPU。llm.npu 构建于 MLLM [23]（一种最先进的移动 LLM 引擎）和 QNN 框架 [25]（高通神经处理 SDK）之上。它支持从 Hugging Face [21] 导出的标准 LLM 格式。为了促进 LLM 执行，除了 QNN 支持的操作之外，我们还实现了 KVCache、SiLU、RMSNorm、ROPE 等特定操作。为了减少 CPU/GPU 和 NPU 之间的上下文切换开销，llm.npu 利用共享缓冲区来同步来自不同处理器的中间结果。对于端到端推理，llm.npu 与任何解码引擎兼容，并利用 MLLM CPU 后端进行解码阶段，易于实现，默认块长度为 256。异常层的默认剪枝率为 85%。llm.npu 在当前实现中不考虑资源检测和争用。

该原型还包含两项优化。(1) 我们广泛的实验表明，移动 NPU 通常更喜欢 CNN 架构中的张量大小（例如，相等的“高度”和“宽度”）。例如，权重为的线性层

2048×2048 对于 1024×1×2048 和 32×32×2048 的输入产生相同的结果，但使用 32×32×2048 可以将执行延迟减少 1.62×。因此，llm.npu 在准备阶段会分析线性层所有可能的等效形状，并选择最有效的形状。(2) 移动 NPU 通常访问有限的内存区域（例如，Hexagon NPU 为 4GB），该内存区域可能小于 LLM 权重的大小。为了在有限的内存内最大化预填充加速，llm.npu 优先在 NPU 上执行计算密集型任务，例如 FFN，以提高效率。

4.1 实验设置

硬件设置。我们在两部配备不同 Qualcomm SoC 的智能手机上测试 llm.npu：Redmi K70 Pro（Snapdragon 8gen3，24GB 内存）和 Redmi K60 Pro（Snapdragon 8gen2，16GB 内存）。所有设备均运行 Android OS 13。

模型和数据集。我们使用各种典型的移动大小的 LLM 进行了测试：Qwen1.5-1.8B [27]、Gemma-2B [9]、Phi2-2.7B [16]、LLaMA2-Chat-7B [11] 和 Mistral-7B [14]。为了评估 llm.npu 的量化精度，我们采用了广泛认可的 LLM 基准，包括 LAMBADA [68]、HellaSwag [103]、Wino Grande [4]、OpenBookQA [62] 和 MMLU [43]。对于推理速度实验，我们选择了来自 Longbench、2wikimqa 和 TriviaQA [28] 的基于检索的数据集，用于模拟上下文感知生成任务，例如自动电子邮件回复。此外，我们使用 DroidTask 数据集 [22] 评估了 llm.npu 的屏幕问答和将指令映射到 UI 操作场景，以模拟基于代理的 UI 自动化任务。

基线。我们主要将 llm.npu 与 5 个基线进行比较，包括 3 个广泛使用的移动 LLM 引擎（TFLite [57]、MNN [49] 和 llama.cpp [59]）。这些引擎仅支持移动 CPU 和 GPU；2 个高级基线是 MLC-LLM [75]（一种用于设备上 GPU 的 LLM 编译器）和 PowerInfer-v2（也利用移动 NPU 来加速预填充 [94]）。由于 PowerInfer-v2 不是开源的，我们使用其论文中报告的数据。需要注意的是，这些基线通常仅支持我们评估的 5 个**大语言模型**。

指标。我们主要测量 LLM 推理精度、预填充延迟、预填充能耗、预填充内存消耗和端到端推理延迟。能耗是通过 Android 操作系统中的 /sys/class/power_supply 每 100ms 进行一次分析来获得的。所有实验重复三次，我们报告平均数。

4.2 预填充性能。

我们在两台设备上以 64、256 和 1024 个令牌的提示长度评估 llm.npu 的预填充性能（速度和能耗），如图 14 和 15 所示。尽管数据集之间存在异常变化，但对预填充性能的总体影响很小，因此我们报告来自 Long-Bench 数据集的结果。结果表明，llm.npu 在这两个指标上始终优于所有基线，并且随着提示长度的增加，优势变得更加明显。

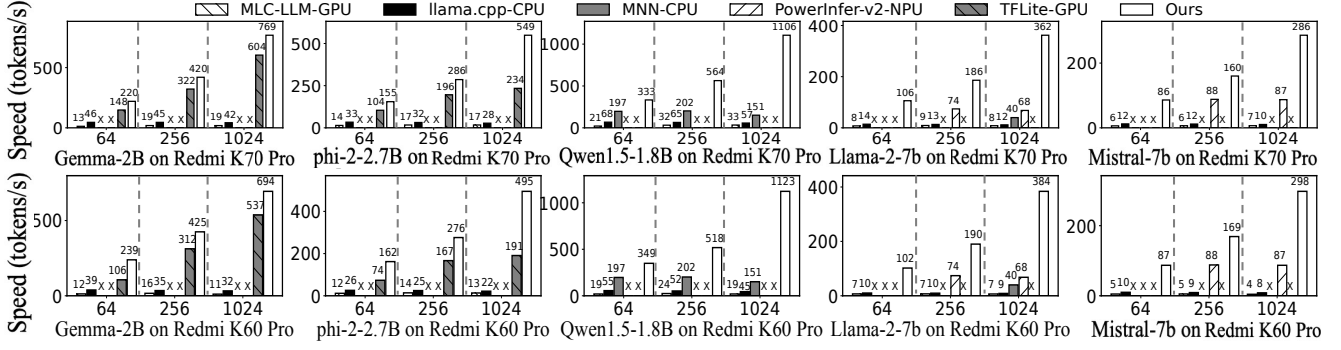


Figure 14. Prefill speed under different prompt lengths on different devices (datasets: Longbench-2wiki-Multi-doc QA).

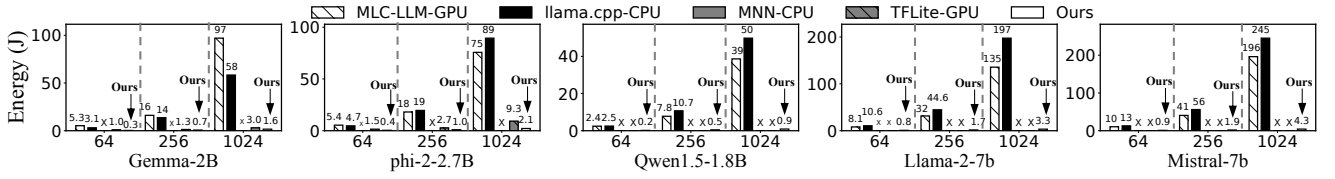


Figure 15. Energy consumption under different prompt lengths on Redmi K60 Pro (datasets: Longbench-2wiki-Multi-doc QA).

Prefill speed. For prompt length of 1024 tokens, 11m.npu can reduce prefill latency by 18.17–38.4×, 7.3×, 32.5–43.6×, and 1.27–2.34× on Redmi K70 Pro compared with llama.cpp-CPU, MNN-CPU, MLC-GPU, TFLite-GPU, respectively. On the Redmi K60 Pro, these improvements are 21.3–41.3×, 7.43×, 37.2–69.3×, and 1.3–2.6×, respectively. These speedups are due to 11m.npu’s use of three innovative techniques that fully leverage mobile NPUs, including *shadow outlier execution*, high-efficiency per-tensor MatMul, and *out-of-order subgraph execution*. Compared with PowerInfer-V2-NPU, a baseline also using NPU for prefilling, 11m.npu can achieve 3.28–5.32× and 3.4–5.6× speedup on two devices, respectively, by employing NPU-friendly INT8 linear computation and fine-grained subgraph scheduling (§3.4).

For prompt length of 64 tokens, the prefill speed of 11m.npu is 14.86–7.10×, 1.69×, 10.91–17.32×, 1.48×, and 1.81–2.51× faster than llama.cpp-CPU, MNN-CPU, MLC-GPU, TFLite-GPU, and PowerInfer-V2-NPU respectively, with speedups averaging 10.5×, 4.31×, 2.68×, 1.02×, and 1.96× lower than those for 1024-token prompts. This is because a shorter prompt can lead to a padding problem and limit 11m.npu’s out-of-order execution scheduling efficiency.

Prefill energy consumption. Energy consumption was evaluated on the Redmi K60 Pro, the only rootable device. PowerInfer-V2 was excluded due to the lack of energy consumption data and open-source code. For 1024-token prompts, 11m.npu reduces energy consumption by 35.63–59.52×, 35.21–59.25×, and 1.85–4.32× compared to llama.cpp-CPU, MLC-GPU, and TFLite-GPU, respectively. For 64-token prompts, the savings are 10.38–14.12×, 10.38–17.79×, and 3.22–3.67×,

respectively. These savings are due to the high energy efficiency of mobile NPUs and 11m.npu’s three novel techniques for maximizing NPU performance. Typically, during the LLM prefill stage, all CPU cores are fully utilized, consuming the highest power; NPUs operate at 500–750 MHz, consuming the least power.

4.3 End-to-end performance

We evaluate the real-world performance of 11m.npu against baseline systems using three workloads: UI automation on DroidTask datasets, context-aware automated email replies on LongBench datasets, chat summaries on Persona-Chat datasets. Table 5 presents the prompt and output lengths of the three datasets, along with their corresponding prefill, decode, and end-to-end inference latency results. End-to-end speedups in Table 5 are the geometric mean of each sample. Our key observation is that **11m.npu always achieves the lowest inference latency across all three datasets.**

For LongBench datasets, 11m.npu shows significant speed improvements: 23.0–46.2× over llama.cpp-CPU, 16.5–36.4× over MLC-LLM-GPU, 4.08–4.19× over MNN-CPU, 3.51–3.73× over PowerInfer-V2-NPU, and 1.27–2.03× over TFLite-GPU. This impressive performance is primarily due to 11m.npu’s superior efficiency during the prefill stage. The speedup against TFLite-GPU is lower since 11m.npu currently relies on a CPU backend for decoding with no optimization, while TFLite utilizes GPU. Notably, 11m.npu is compatible with any decoding engine, which means once TFLite is open-sourced, 11m.npu can integrate it as the decoding backend, potentially enhancing performance further.

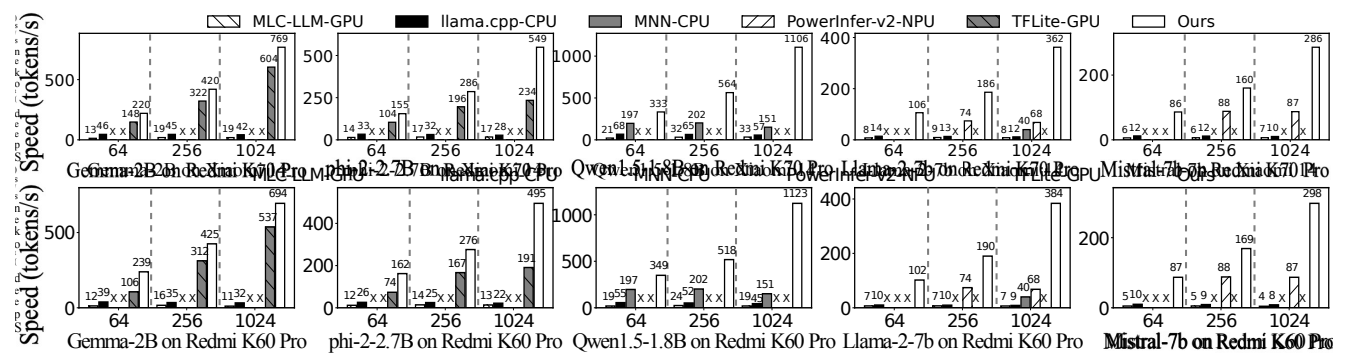


图 14. 不同设备上不同提示长度下的预填充速度（数据集：Longbench-2wiki-Multi-doc QA）。

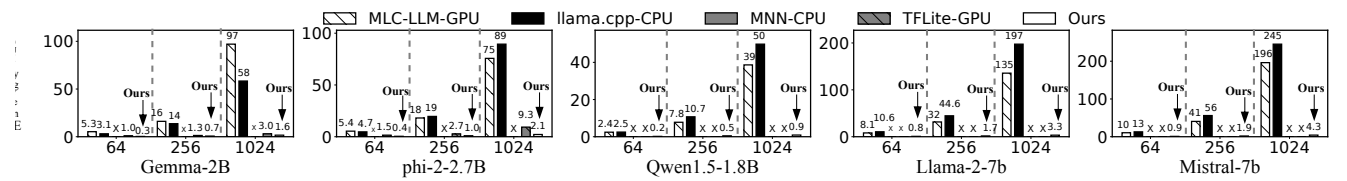


图15 Redmi K60 Pro不同提示时长下的能耗（数据集：Longbench-2wiki-Multi-doc Q 一个）。

预充速度。对于 1024 个令牌的提示长度，与 llama.cpp-CPU、MNN-CPU、MLC-GPU、TFLite-GPU 相比，llm.npu 在 Redmi K70 Pro 上可分别将预填充延迟减少 18.17–38.4×、7.3×、32.5–43.6× 和 1.27–2.34×。在 Redmi K60 Pro 上，这些改进分别为 21.3–41.3×、7.43×、37.2–69.3× 和 1.3–2.6×。这些加速归功于 llm.npu 使用了三种充分利用移动 NPU 的创新技术，包括阴影离群值执行、高效每张量 MatMul 和乱序子图执行。与同样使用 NPU 进行预填充的基线 PowerInfer-V2-NPU 相比，llm.npu 通过采用 NPU 友好的 INT8 线性计算和细粒度子图调度 (§3.4)，可以在两个设备上分别实现 3.28–5.32× 和 3.4–5.6× 加速。

对于 64 个 token 的提示长度，llm.npu 的预填充速度比 llama.cpp-CPU、MNN-CPU、MLC-GPU 快 14.86–7.10×、1.69×、10.91–17.32×、1.48× 和 1.81–2.51×，TFLite-GPU 和 PowerInfer-V2-NPU 的平均加速比 1024 个令牌提示低 10.5×、4.31×、2.68×、1.02× 和 1.96×。这是因为较短的提示可能会导致填充问题并限制 llm.npu 的乱序执行调度效率。

预填充能量消耗。能耗评估是在唯一可 Root 的设备 Redmi K60 Pro 上进行的。由于缺乏能耗数据和开源代码，PowerInfer-V2 被排除在外。对于 1024 个令牌提示，与 llama.cpp-CPU、MLC-GPU 和 TFLite-GPU 相比，llm.npu 分别将能耗降低了 35.63–59.52×、35.21–59.25× 和 1.85–4.32×。对于 64 个令牌提示，节省的费用为 10.38–14.12×、10.38–17.79× 和 3.22–3.67×，

分别。这些节省归功于移动 NPU 的高能效以及 llm.npu 的三项新技术，可最大限度地提高 NPU 性能。通常，在 LLM 预填充阶段，所有 CPU 核心都被充分利用，消耗的功率最高；NPU 的运行频率为 500–750 MHz，功耗最低。

4.3 端到端性能

我们使用三种工作负载根据基准系统评估 llm.npu 的实际性能：DroidTask 数据集上的 UI 自动化、LongBench 数据集上的上下文感知自动电子邮件回复、Persona-Chat 数据集上的聊天摘要。表 5 显示了三个数据集的提示和输出长度，以及相应的预填充、解码和端到端推理延迟结果。表 5 中的端到端加速是每个样本的几何平均值。我们的主要观察结果是 llm.npu 始终在所有三个数据集上实现最低的推理延迟。

对于 LongBench 数据集，llm.npu 显示出显著的速度改进：比 llama.cpp-CPU 提高 23.0–46.2×，比 MLC-LLM-GPU 提高 16.5–36.4×，比 MNN-CPU 提高 4.08–4.19×，比 PowerInfer-V2-NPU 提高 3.51–3.73×，以及 1.27–2.03× 通过 TFLite-GPU。这种令人印象深刻的表现主要归功于 llm.npu 在预填充阶段的卓越效率。相对于 TFLite-GPU 的加速比较低，因为 llm.npu 目前依赖 CPU 后端进行解码，没有优化，而 TFLite 使用 GPU。值得注意的是，llm.npu 与任何解码引擎兼容，这意味着一旦 TFLite 开源，llm.npu 就可以将其集成为解码后端，从而有可能进一步提高性能。

Table 5. End-to-end latency comparison using real mobile applications execution on Redmi K70 Pro. Each cell is organized in the following format: end-to-end latency (prefill latency, decode latency). Unit: seconds.

LLM	Datasets	MLC	LCPP	MNN	PI	TFLite	Ours	Speedup
Qwen1.5-1.8B	Longbench: 2wiki-Multi-doc QA prompt length: 1451–1672 tokens output length: 2–4 tokens	45.6 (45.43, 0.17)	26.7 (26.43, 0.24)	10.6 (10.01, 0.59)	-	-	1.7 (1.49, 0.24)	6.2-26.8×
Gemma-2B		78.4 (78.03, 0.32)	34.6 (34.31, 0.25)	-	-	2.6 (2.40, 0.19)	1.9 (1.68, 0.25)	1.4-41.3×
Phi-2-2.7B		87 (86.70, 0.28)	53.3 (53.00, 0.34)	13 (11.98, 1.05)	-	6.3 (6.06, 0.28)	3.1 (2.79, 0.34)	2.0-28.1×
LlaMA-2-7B		184.7 (184.20, 0.45)	146 (145.25, 0.70)	22.4 (22.35, 0.05)	19.8 (19.04, 0.78)	-	5.3 (4.62, 0.70)	3.7-34.8×
Mistral-7b		254.2 (253.45, 0.72)	200.2 (199.37, 0.87)	-	20 (19.22, 0.78)	-	5.5 (4.61, 0.87)	3.6-46.2×
Geo-mean (speedup)		34.7×	21.8×	4.8×	3.7×	1.7×	-	
Qwen1.5-1.8B	Longbench: TriviaQA prompt length: 1511–1787 tokens output length: 5–11 tokens)	46 (45.68)	27 (26.57, 0.47)	11.2 (10.07, 1.15)	-	-	2.0 (1.49, 0.47)	5.6-23.0×
Gemma-2B		81.8 (81.25, 0.26)	36.2 (35.72, 0.43)	-	-	2.8 (2.50, 0.32)	2.2 (1.75, 0.43)	1.3-37.2×
Phi-2-2.7B		91.4 (90.94, 0.50)	56.3 (55.60, 0.68)	14.7 (12.57, 2.11)	-	6.8 (6.36, 0.40)	3.6 (2.94, 0.68)	1.9-25.4×
LlaMA-2-7B		197.3 (196.43, 0.83)	156.2 (154.90, 1.28)	23.8 (20.83, 3.00)	21.8 (20.31, 1.34)	-	6.2 (4.93, 1.28)	3.5-31.8×
Mistral-7b		266.2 (264.93, 1.31)	210 (208.41, 1.57)	-	21.5 (20.10, 0.41)	-	6.4 (4.82, 1.57)	3.4-41.6×
Geo-mean (speedup)		31.0×	19.6×	4.4×	3.4×	1.6×	-	
Qwen1.5-1.8B	DroidTask: clock prompt length: 656–827 tokens output length: 1–5 tokens)	21 (20.84, 0.14)	10.4 (10.16, 0.25)	3.9 (3.25, 0.61)	-	-	1.4 (1.16, 0.25)	2.80-15×
Gemma-2B		39.4 (39.11, 0.25)	16.5 (16.25, 0.21)	-	-	2.5 (2.30, 0.18)	1.2 (0.97, 0.23)	2.1-32.8×
Phi-2-2.7B		46.6 (46.40, 0.24)	25 (24.62, 0.34)	7.4 (6.34, 1.06)	-	4.2 (3.98, 0.2)	3.1 (2.72, 0.34)	1.4-15.0×
LlaMA-2-7B		87.7 (87.38, 0.33)	60.4 (59.83, 0.63)	10.6 (8.58, 2.11)	11.1 (10.74, 0.37)	-	4.8 (4.28, 0.53)	2.2-18.3×
Mistral-7b		122.3 (121.82, 0.51)	68.6 (67.91, 0.65)	-	12 (11.64, 0.37)	-	4.9 (4.26, 0.65)	2.4-25.0×
Geo-mean (speedup)		20.2×	10.8×	2.4×	2.4×	1.7×	-	
Qwen1.5-1.8B	DroidTask: clock prompt length: 505–645 tokens output length: 3–5 tokens	16.2 (16.05, 0.14)	8.1 (7.83, 0.25)	3.1 (2.5, 0.61)	-	-	1.1 (0.90, 0.25)	2.8-14.7×
Gemma-2B		29.4 (29.14, 0.52)	12.3 (12.11, 0.20)	-	-	1.9 (1.71, 0.18)	0.9 (0.72, 0.23)	2.1-32.7×
Phi-2-2.7B		35.4 (35.15, 0.23)	19 (18.65, 0.34)	5.9 (4.80, 1.05)	-	3.2 (3.01, 1.2)	2.4 (2.07, 0.34)	1.3-14.8×
LlaMA-2-7B		63.7 (63.38, 0.33)	43.9 (43.40, 0.52)	7.7 (5.06, 1.64)	8.2 (7.79, 0.37)	-	3.6 (3.10, 0.52)	2.1-17.7×
Mistral-7b		90.1 (89.60, 0.51)	50.6 (49.95, 0.65)	-	8.9 (8.56, 0.37)	-	3.8 (3.13, 0.65)	2.3-23.7×
Geo-mean (speedup)		19.7×	10.5×	2.5×	2.3×	1.7×	-	
Qwen1.5-1.8B	Persona-Chat prompt length: 488–584 tokens output length: 35–57 tokens	18.74 (16.54, 2.20)	11.86 (8.07, 3.79)	11.87 (2.58, 9.29)	-	-	6.72 (2.92, 3.80)	1.8-2.8×
Gemma-2B		30.9 (27.05, 3.82)	14.3 (11.24, 3.02)	-	-	4.3 (1.59, 2.67)	4.2 (0.67, 3.48)	1.02-7.4×
Phi-2-2.7B		36.1 (32.51, 3.57)	22.5 (15.24, 3.22)	20.6 (4.43, 16.19)	-	5.9 (2.79, 3.06)	5.1 (1.91, 3.22)	1.2-7.1×
LlaMA-2-7B		64.2 (58.96, 5.20)	48.7 (40.37, 8.30)	45.5 (7.15, 38.23)	13.0 (7.25, 5.77)	-	11.2 (2.88, 8.30)	1.2-5.7×
Mistral-7b		90.5 (82.54, 8.0)	56.3 (46.02, 10.28)	-	13.7 (7.89, 5.76)	-	13.2 (2.88, 10.28)	1.04-6.9×
Geo-mean (speedup)		10.5×	3.5×	3.1×	1.1×	1.1×	-	

*LCPP and PI in the first row represent llama.cpp and PowerInfer-V2, respectively.

For the DroidTask datasets, 11m.npu reduces end-to-end inference latency by 7.9–12.5×

compared to MLC-LLM-GPU, 2.38–2.45×

compared to MNN-CPU, 2.27–2.44×

compared to PowerInfer-V2-NPU, and 1.35–2.38×

compared to TFLite-GPU. The performance gains are slightly smaller for DroidTask datasets due to shorter prompts in UI automation versus email writing.

表 5. 在 Redmi K70 Pro 上使用真实移动应用程序执行进行的端到端延迟比较。每个单元按以下格式组织：端到端延迟（预填充延迟、解码延迟）。单位：秒。

LLM	Datasets	MLC	LCPP	MNN	PI	TFLite	Ours	Speedup
Qwen1.5-1.8B	Longbench: 2wiki-Multi-doc QA prompt length: 1451–1672 tokens output length: 2–4 tokens	45.6 (45.43, 0.17)	26.7 (26.43, 0.24)	10.6 (10.01, 0.59)	-	-	1.7 (1.49, 0.24)	6.2-26.8×
Gemma-2B		78.4 (78.03, 0.32)	34.6 (34.31, 0.25)	-	-	2.6 (2.40, 0.19)	1.9 (1.68, 0.25)	1.4-41.3×
Phi-2-2.7B		87 (86.70, 0.28)	53.3 (53.00, 0.34)	13 (11.98, 1.05)	-	6.3 (6.06, 0.28)	3.1 (2.79, 0.34)	2.0-28.1×
LlaMA-2-7B		184.7 (184.20, 0.45)	146 (145.25, 0.70)	22.4 (22.35, 0.05)	19.8 (19.04, 0.78)	-	5.3 (4.62, 0.70)	3.7-34.8×
Mistral-7b		254.2 (253.45, 0.72)	200.2 (199.37, 0.87)	-	20 (19.22, 0.78)	-	5.5 (4.61, 0.87)	3.6-46.2×
Geo-mean (speedup)		34.7×	21.8×	4.8×	3.7×	1.7×	-	
Qwen1.5-1.8B	Longbench: TriviaQA prompt length: 1511–1787 tokens output length: 5–11 tokens)	46 (45.68)	27 (26.57, 0.47)	11.2 (10.07, 1.15)	-	-	2.0 (1.49, 0.47)	5.6-23.0×
Gemma-2B		81.8 (81.25, 0.26)	36.2 (35.72, 0.43)	-	-	2.8 (2.50, 0.32)	2.2 (1.75, 0.43)	1.3-37.2×
Phi-2-2.7B		91.4 (90.94, 0.50)	56.3 (55.60, 0.68)	14.7 (12.57, 2.11)	-	6.8 (6.36, 0.40)	3.6 (2.94, 0.68)	1.9-25.4×
LlaMA-2-7B		197.3 (196.43, 0.83)	156.2 (154.90, 1.28)	23.8 (20.83, 3.00)	21.8 (20.31, 1.34)	-	6.2 (4.93, 1.28)	3.5-31.8×
Mistral-7b		266.2 (264.93, 1.31)	210 (208.41, 1.57)	-	21.5 (20.10, 0.41)	-	6.4 (4.82, 1.57)	3.4-41.6×
Geo-mean (speedup)		31.0×	19.6×	4.4×	3.4×	1.6×	-	
Qwen1.5-1.8B	DroidTask: clock prompt length: 656–827 tokens output length: 1–5 tokens)	21 (20.84, 0.14)	10.4 (10.16, 0.25)	3.9 (3.25, 0.61)	-	-	1.4 (1.16, 0.25)	2.80-15×
Gemma-2B		39.4 (39.11, 0.25)	16.5 (16.25, 0.21)	-	-	2.5 (2.30, 0.18)	1.2 (0.97, 0.23)	2.1-32.8×
Phi-2-2.7B		46.6 (46.40, 0.24)	25 (24.62, 0.34)	7.4 (6.34, 1.06)	-	4.2 (3.98, 0.2)	3.1 (2.72, 0.34)	1.4-15.0×
LlaMA-2-7B		87.7 (87.38, 0.33)	60.4 (59.83, 0.63)	10.6 (8.58, 2.11)	11.1 (10.74, 0.37)	-	4.8 (4.28, 0.53)	2.2-18.3×
Mistral-7b		122.3 (121.82, 0.51)	68.6 (67.91, 0.65)	-	12 (11.64, 0.37)	-	4.9 (4.26, 0.65)	2.4-25.0×
Geo-mean (speedup)		20.2×	10.8×	2.4×	2.4×	1.7×	-	
Qwen1.5-1.8B	DroidTask: clock prompt length: 505–645 tokens output length: 3–5 tokens)	16.2 (16.05, 0.14)	8.1 (7.83, 0.25)	3.1 (2.5, 0.61)	-	-	1.1 (0.90, 0.25)	2.8-14.7×
Gemma-2B		29.4 (29.14, 0.52)	12.3 (12.11, 0.20)	-	-	1.9 (1.71, 0.18)	0.9 (0.72, 0.23)	2.1-32.7×
Phi-2-2.7B		35.4 (35.15, 0.23)	19 (18.65, 0.34)	5.9 (4.80, 1.05)	-	3.2 (3.01, 1.2)	2.4 (2.07, 0.34)	1.3-14.8×
LlaMA-2-7B		63.7 (63.38, 0.33)	43.9 (43.40, 0.52)	7.7 (5.06, 1.64)	8.2 (7.79, 0.37)	-	3.6 (3.10, 0.52)	2.1-17.7×
Mistral-7b		90.1 (89.60, 0.51)	50.6 (49.95, 0.65)	-	8.9 (8.56, 0.37)	-	3.8 (3.13, 0.65)	2.3-23.7×
Geo-mean (speedup)		19.7×	10.5×	2.5×	2.3×	1.7×	-	
Qwen1.5-1.8B	Persona-Chat prompt length: 488–584 tokens output length: 35–57 tokens)	18.74 (16.54, 2.20)	11.86 (8.07, 3.79)	11.87 (2.58, 9.29)	-	-	6.72 (2.92, 3.80)	1.8-2.8×
Gemma-2B		30.9 (27.05, 3.82)	14.3 (11.24, 3.02)	-	-	4.3 (1.59, 2.67)	4.2 (0.67, 3.48)	1.02-7.4×
Phi-2-2.7B		36.1 (32.51, 3.57)	22.5 (15.24, 3.22)	20.6 (4.43, 16.19)	-	5.9 (2.79, 3.06)	5.1 (1.91, 3.22)	1.2-7.1×
LlaMA-2-7B		64.2 (58.96, 5.20)	48.7 (40.37, 8.30)	45.5 (7.15, 38.23)	13.0 (7.25, 5.77)	-	11.2 (2.88, 8.30)	1.2-5.7×
Mistral-7b		90.5 (82.54, 8.0)	56.3 (46.02, 10.28)	-	13.7 (7.89, 5.76)	-	13.2 (2.88, 10.28)	1.04-6.9×
Geo-mean (speedup)		10.5×	3.5×	3.1×	1.1×	1.1×	-	

*第一行中的LCPP和PI分别代表llama.cpp和PowerInfer-V2。

对于 DroidTask 数据集，与 llama.cpp-CPU 相比，llm.npu 将端到端推理延迟减少了 7.9–12.5%，

与 MLC-LLM-GPU 相比为 15.0–32.8%，与 MNN-CPU 相比为 2.38–2.45%，与 PowerInfer-V2-NPU 相比为 2.27–2.44%，与 TFLite-GPU 相比为 1.35–2.38%。由于 UI 自动化中的提示与电子邮件编写相比较短，DroidTask 数据集的性能提升略小。

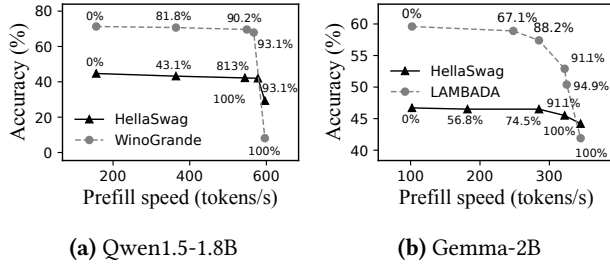


Figure 16. Generation speed and accuracy across various outlier pruning rates. The text positioned above or below the plots represents the corresponding pruning rates.

For the Persona-Chat datasets, `llm.npu` reduces end-to-end inference latency by an average of $3.5\times$ compared to `llama.cpp-CPU`, $10.5\times$ compared to `MLC-LLM-GPU`, $3.1\times$ compared to `MNN-CPU`, 1.1 times compared to `PowerInfer-V2-NPU`, and $1.1\times$ compared to `TFLite-GPU`. The primary reason for the reduced performance improvement relative to other datasets is the larger number of output tokens generated by this dataset. Since `llm.npu` currently supports only CPU-based decoding, this slower decoding backend results in increased end-to-end latency. This issue can be mitigated by utilizing a mobile GPU as the decoding backend (see Section 4.6 for details).

4.4 Inference accuracy

Overall accuracy. We investigate the inference accuracy of `llm.npu` on 5 LLM benchmarks: LAMBADA [68], MMLU [43], WinoGrande [4], OpenBookQA [62] and HellaSwag [103]. For comparison, we evaluated 4 alternatives: FP16 (non-quantization), K-Quant [59] (used in `llama.cpp`), SmoothQuant (state-of-the-art per-tensor method) [84], and LLM.Int8() (state-of-the-art float outlier handling method) [33]. **`llm.npu` achieves negligible accuracy loss, and significantly outperforms the other quantization algorithms**, as shown in Table 6.

Specifically, `llm.npu` is, on average, only 1% less accurate than FP16, and it shows an accuracy improvement of up to 32.9% over SmoothQuant and up to 70.9% over K-Quant. This improvement over SmoothQuant, which uses static profiling to smooth outliers to normal values, is due to `llm.npu`'s dynamic handling of outlier positions with CPU float precision. `llm.npu` addresses outliers at the element level, providing higher precision than K-Quant that uses group-level quantization scales. Furthermore, `llm.npu` achieves comparable accuracy (0.1% average loss) to LLM.Int8(), as both handle outliers with float precision. But `llm.npu` better utilizes NPU-specific computational features, maintaining high accuracy and NPU efficiency.

Trade-offs between accuracy and performance. `llm.npu` balances generation accuracy and speed by adjusting outlier pruning rates, **with higher pruning rates leading to**

Table 6. LLM capability accuracy on `llm.npu` and baselines. "SQ": SmoothQuant; "Int8()": LLM.Int8(); "Degrad.": accuracy degradation compared to FP16.

LAMBADA	FP16	SQ	INT8()	K-Quant	Ours	Ours Degrad.
Qwen1.5-1.8B	71.1%	65.6%	71.0%	62.7%	71.7%	+0.6%
Gemma2-2B	59.6%	45.8%	59.2%	56.9%	59.4%	-0.2%
Phi-2-2.7B	72.2%	66.1%	71.7%	59.3%	67.5%	-4.7%
LlaMA-2-7B	87.5%	71.9%	88.0%	15.6%	86.3%	-1.2%
Mistral-7b	84.8%	51.2%	85.3%	23.9%	84.1%	-0.7%
Avg. Degrad.	-	-14.9%	0%	-31.3%	-1.2%	

HellaSwag	FP16	SQ	INT8()	K-Quant	Ours	Ours Degrad.
Qwen1.5-1.8B	43.8%	40.9%	43.5%	44.3%	43.8%	0%
Gemma2-2B	46.5%	43.8%	46.1%	45.4%	47.3%	+0.8%
Phi-2-2.7B	48.2%	46.2%	47.7%	47.6%	46.9%	-1.3%
LlaMA-2-7B	52.8%	44.4%	53.1%	50.5%	53.5%	+0.7%
Mistral-7b	57.4%	44.9%	57.9%	57.0%	57.0%	-0.4%
Avg. Degrad.	-	-5.7%	-0.1%	-0.8%	-0.0%	

WinoGrande	FP16	SQ	INT8()	K-Quant	ours	Ours Degrad.
Qwen1.5-1.8B	58.3%	51.0%	58.2%	59.0%	59.3%	+1.0%
Gemma2-2B	58.3%	54.8%	59.0%	58.5%	59.5%	+1.2%
Phi-2-2.7B	72.2%	68.9%	72.4%	72.5%	70.2%	-2.0%
LlaMA-2-7B	65.2%	56.9%	66.2%	67.4%	65.1%	-0.1%
Mistral-7b	73.5%	59.1%	73.3%	73.5%	73.1%	-0.4%
Avg. Degrad.	-	-7.4%	+0.3%	+0.7%	-0.1%	

OpenBookQA	FP16	SQ	INT8()	K-Quant	ours	Ours Degrad.
Qwen1.5-1.8B	28.8%	23.0%	28.5%	28.0%	26.6%	-2.2%
Gemma2-2B	33.7%	28.0%	34.2%	33.0%	38.4%	+4.7%
Phi-2-2.7B	41.0%	35.9%	40.2%	39.5%	37.7%	-3.3%
LlaMA-2-7B	32.7%	25.0%	32.0%	31.5%	31.1%	-1.6%
Mistral-7b	39.4%	25.6%	39.3%	37.9%	39.3%	-0.1%
Avg. Degrad.	-	-7.6%	-0.3%	-1.1%	-0.5%	

MMLU	FP16	SQ	INT8()	K-Quant	ours	Ours Degrad.
Qwen1.5-1.8B	29.7%	27.9%	29.1%	29.8%	30.8%	+1.1%
Gemma2-2B	35.7%	32.1%	35.1%	35.1%	36.4%	+0.7%
Phi-2-2.7B	35.4%	35.3%	35.6%	35.7%	36.7%	+1.3%
LlaMA-2-7B	37.8%	29.2%	38.1%	34.4%	36.9%	-0.9%
Mistral-7b	42.1%	30.9%	41.4%	42.3%	41.0%	-1.1%
Avg. Degrad.	-	-5.1%	-0.3%	-0.7%	+0.2%	

faster generation but lower quality. To comprehensively show how outlier pruning rates influence generate accuracy and speed, we conducted experiments using Qwen1.5-1.8B and Gemma-2B on the Redmi K70 Pro with the HellaSwag [103] and LAMBADA [68] datasets. The results are shown in Figure 16.

When no outliers are pruned (pruning rate = 0%), `llm.npu` achieves the highest accuracy, albeit with the slowest generation speed. Specifically, the Qwen1.5-1.8B model can attain an accuracy of 71.3% on the Lambada dataset, with a corresponding generation speed of 156 tokens per second. Conversely, the Gemma-2B model demonstrates an accuracy of 59.6% and a generation speed of 102 tokens per second. Upon pruning 80% of outliers, the Qwen1.5-1.8B model's accuracy decreases to 42.2%, while its generation speed increases to 544 tokens per second. Similarly, the Gemma-2B model's accuracy declines to 57.4%, yet its generation speed rises to 285

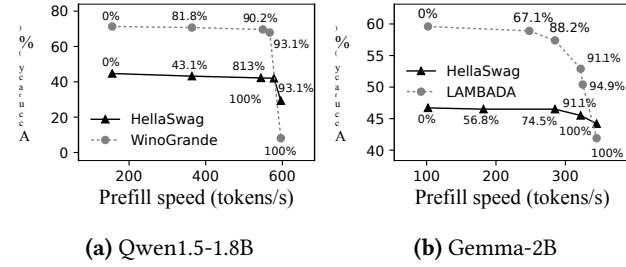


图 16. 不同异常值修剪率的生成速度和准确性。位于图上方或下方的文本表示相应的修剪率。

对于 Persona-Chat 数据集，与 llama.cpp-CPU 相比，llm.npu 将端到端推理延迟平均降低了 3.5 $\times$ ，与 MLC-LLM-GPU 相比，平均降低了 10.5 $\times$ ，与 MNN-CPU 相比，降低了 3.1 $\times$ ，与 PowerInfer-V2-NPU 相比，降低了 1.1 倍，与 TFLite-GPU 相比，降低了 1.1 $\times$ 。相对于其他数据集，性能改进下降的主要原因是该数据集生成的输出标记数量较多。由于 llm.npu 目前仅支持基于 CPU 的解码，因此较慢的解码后端会导致端到端延迟增加。这个问题可以通过使用移动 GPU 作为解码后端来缓解（详细信息请参见第 4.6 节）。

4.4 推理精度

总体准确度。我们研究了 llm.npu 在 5 个 LLM 基准上的推理准确性：LAMBADA [68]、MMLU [43]、WinoGrande [4]、OpenBookQA [62] 和 HellaSwag [103]。为了进行比较，我们评估了 4 个替代方案：FP16（非量化）、K-Quant [59]（在 llama.cpp 中使用）、SmoothQuant（最先进的每张量方法）[84] 和 LLM.Int8()（最先进的浮点异常值处理方法）[33]。llm.npu 的精度损失可以忽略不计，并且显著优于其他量化算法，如表 6 所示。

具体来说，llm.npu 的平均准确度仅比 FP16 低 1%，但与 SmoothQuant 相比，准确度提高了 32.9%，比 K-Quant 提高了 70.9%。SmoothQuant 使用静态分析将离群值平滑至正常值，这一改进归功于 llm.npu 以 CPU 浮点精度动态处理离群值位置。llm.npu 在元素级别解决异常值，提供比使用组级别量化尺度的 K-Quant 更高的精度。此外，llm.npu 实现了与 LLM.Int8() 相当的精度（平均损失 0.1%），因为两者都以浮点精度处理异常值。但 llm.npu 更好地利用了 NPU 特定的计算特性，保持了高精度和 NPU 效率。

准确性和性能之间的权衡。llm.npu 通过调整离群剪枝率来平衡生成精度和速度，较高的剪枝率会导致

表 6. llm.npu 和基准上的 LLM 能力准确性。“SQ”：SmoothQuant；“Int8()”：LLM.Int8()；“Degrad.”：与 FP16 相比，精度下降。

LAMBADA	FP16	SQ	INT8()	K-Quant	Ours	Ours Degrad.
Qwen1.5-1.8B	71.1%	65.6%	71.0%	62.7%	71.7%	+0.6%
Gemma2-2B	59.6%	45.8%	59.2%	56.9%	59.4%	-0.2%
Phi-2-2.7B	72.2%	66.1%	71.7%	59.3%	67.5%	-4.7%
LlaMA-2-7B	87.5%	71.9%	88.0%	15.6%	86.3%	-1.2%
Mistral-7b	84.8%	51.2%	85.3%	23.9%	84.1%	-0.7%
Avg. Degrad.	-	-14.9%	0%	-31.3%	-1.2%	

HellaSwag	FP16	SQ	INT8()	K-Quant	Ours	Ours Degrad.
Qwen1.5-1.8B	43.8%	40.9%	43.5%	44.3%	43.8%	0%
Gemma2-2B	46.5%	43.8%	46.1%	45.4%	47.3%	+0.8%
Phi-2-2.7B	48.2%	46.2%	47.7%	47.6%	46.9%	-1.3%
LlaMA-2-7B	52.8%	44.4%	53.1%	50.5%	53.5%	+0.7%
Mistral-7b	57.4%	44.9%	57.9%	57.0%	57.0%	-0.4%
Avg. Degrad.	-	-5.7%	-0.1%	-0.8%	-0.0%	

WinoGrande	FP16	SQ	INT8()	K-Quant	ours	Ours Degrad.
Qwen1.5-1.8B	58.3%	51.0%	58.2%	59.0%	59.3%	+1.0%
Gemma2-2B	58.3%	54.8%	59.0%	58.5%	59.5%	+1.2%
Phi-2-2.7B	72.2%	68.9%	72.4%	72.5%	70.2%	-2.0%
LlaMA-2-7B	65.2%	56.9%	66.2%	67.4%	65.1%	-0.1%
Mistral-7b	73.5%	59.1%	73.3%	73.5%	73.1%	-0.4%
Avg. Degrad.	-	-7.4%	+0.3%	+0.7%	-0.1%	

OpenBookQA	FP16	SQ	INT8()	K-Quant	ours	Ours Degrad.
Qwen1.5-1.8B	28.8%	23.0%	28.5%	28.0%	26.6%	-2.2%
Gemma2-2B	33.7%	28.0%	34.2%	33.0%	38.4%	+4.7%
Phi-2-2.7B	41.0%	35.9%	40.2%	39.5%	37.7%	-3.3%
LlaMA-2-7B	32.7%	25.0%	32.0%	31.5%	31.1%	-1.6%
Mistral-7b	39.4%	25.6%	39.3%	37.9%	39.3%	-0.1%
Avg. Degrad.	-	-7.6%	-0.3%	-1.1%	-0.5%	

MMLU	FP16	SQ	INT8()	K-Quant	ours	Ours Degrad.
Qwen1.5-1.8B	29.7%	27.9%	29.1%	29.8%	30.8%	+1.1%
Gemma2-2B	35.7%	32.1%	35.1%	35.1%	36.4%	+0.7%
Phi-2-2.7B	35.4%	35.3%	35.6%	35.7%	36.7%	+1.3%
LlaMA-2-7B	37.8%	29.2%	38.1%	34.4%	36.9%	-0.9%
Mistral-7b	42.1%	30.9%	41.4%	42.3%	41.0%	-1.1%
Avg. Degrad.	-	-5.1%	-0.3%	-0.7%	+0.2%	

生成速度更快，但质量较低。为了全面展示离群剪枝率如何影响生成精度和速度，我们在 Redmi K70 Pro 上使用 Qwen1.5-1.8B 和 Gemma-2B 以及 HellaSwag [103] 和 LAMBADA [68] 数据集进行了实验。结果如图 16 所示。

当没有异常值被剪枝时（剪枝率=0%），llm.npu 实现了最高的准确率，尽管生成速度最慢。具体来说，Qwen1.5-1.8B 模型在 Lambada 数据集上可以达到 71.3% 的准确率，相应的生成速度为每秒 156 个令牌。相反，Gemma-2B 模型的准确度为 59.6%，生成速度为每秒 102 个令牌。在修剪 80% 的异常值后，Qwen1.5-1.8B 模型的准确率下降至 42.2%，而其生成速度则增加至每秒 54 4 个令牌。同样，Gemma-2B 模型的准确率下降至 57.4%，但其生成速度却上升至 285

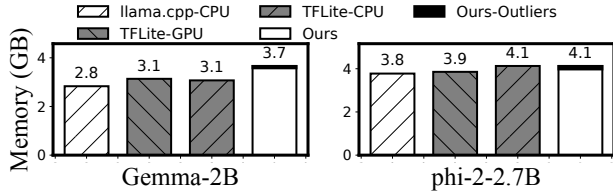
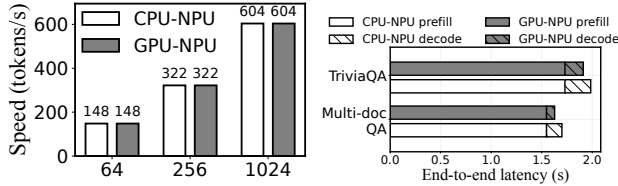


Figure 17. Memory consumption of different baselines (prompt length=512).



(a) Prefill speed under different prompt length **(b)** End-to-end latency on the Longbench dataset

Figure 18. Prefill speed and end-to-end latency comparison between CPU-NPU coordination and GPU-NPU coordination.

tokens per second. When all outliers are pruned, generation accuracy reaches its lowest point, while generation speed is at its highest. Specifically, the Qwen1.5-1.8B model demonstrates an accuracy of merely 8.1% with a generation speed of 596 tokens/s; the Gemma-2B model reports an accuracy of 41.9% alongside a generation speed of 345 tokens/s.

4.5 Memory consumption

We compare 11m.npu with INT8 weight baselines, as mobile NPUs only support INT8 weight computations. Memory consumption results on the Redmi K60 Pro, using a 512-token prompt, are presented in Figure 17. 11m.npu consumes up to $1.32 \times$ more memory than llama.cpp and TFLite. The overhead is due to the MLLM and QNN frameworks, which allocate independent activation buffers for each operator to enhance speed. The tiny additional memory overhead introduced by 11m.npu is its §3.3 *shadow outlier execution* technique (in black), which loads tiny float weights into memory, accounting for only 0.6%–1% of the total memory.

4.6 GPU-NPU coordination analysis

11m.npu is compatible with mobile GPUs; however, we have currently implemented only the CPU-NPU version due to easy implementation. To illustrate the potential advantages of employing a mobile GPU, we simulate GPU-NPU co-scheduling, assuming TFLite serves as the GPU backend. The evaluation is conducted using the Gemma-2B model across varying prompt lengths and the Longbench dataset, specifically on the Redmi K70 Pro, as shown in Figure 18. Our observations indicate that **while GPU-NPU coordination**

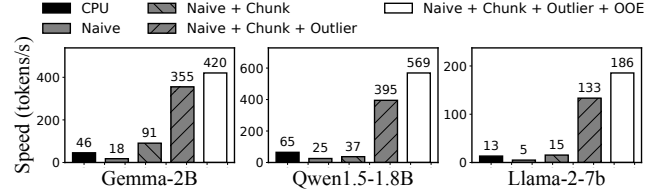


Figure 19. Ablation study of 11m.npu (prompt length=512).

does not enhance the prefilling speed, it effectively reduces the end-to-end latency.

As illustrated in Figure 18(a), the prefill speed achieved through GPU-NPU coordination is equivalent to that of CPU-NPU coordination. This phenomenon occurs because, within the design of 11m.npu, the computational demands placed on mobile CPUs and GPUs are significantly lower than those on the NPU (details in Section 3.4). Consequently, the computational latency of mobile CPUs and GPUs can be effectively hidden by the NPU computation, indicating that the choice of a mobile CPU or mobile GPU is not essential. On the other hand, GPU-NPU coordination can decrease end-to-end latency by 80–90 ms compared to CPU-NPU coordination. This improvement is attributed to the faster decoding speed of GPUs relative to CPUs.

4.7 Ablation study.

We conduct a comprehensive breakdown analysis of the benefits brought by each of 11m.npu's techniques using the Qwen1.5-1.8B, Gemma-2B, and LLaMA2-7B models, as shown in Figure 19. The leftmost bar represents the prefill speed on the CPU using llama.cpp. The second bar shows a naive implementation on mobile NPUs, followed by the incorporation of our three techniques, with the rightmost bar representing 11m.npu. The three techniques are represented by *chunk* (§3.2), *outlier* (§3.3), and *OOE* (§3.4), respectively. We observe that **11m.npu's three techniques make a significant contribution to the overall improvement.**

Firstly, directly offloading LLM prefilling workloads to mobile NPUs results in 2.55–2.68 $\times$ delays, due to the substantial gap between LLMs and mobile NPUs, as detailed in §2.3. Additionally, *chunk-sharing graph* improves prefill speed by 1.46–5.09 $\times$ by reducing graph building and optimization delays. The Gemma-2B model achieves the highest speedup, as it requires more time to build and optimize. Furthermore, *shadow outlier execution* reduces prefill latency by 3.91–8.68 $\times$, allowing per-tensor MatMul operations to fully utilize NPUs with minimal CPU/GPU overhead. Lastly, the *out-of-order subgraph execution* reduces prefill latency by 18%–44% by reducing execution bubbles within the NPU.

5 Discussion and Future Work

Advantages of NPUs over GPUs. Typically, both mobile GPUs and NPUs serve as hardware accelerators for on-device

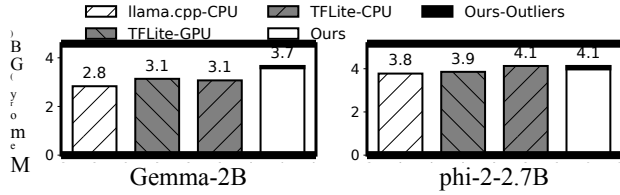


图 17. 不同基线的内存消耗（提示长度=512）。

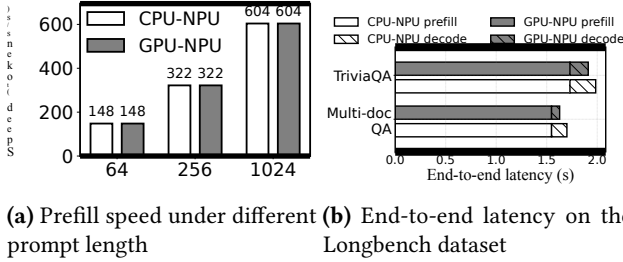


图 18. CPU-NPU 协调和 GPU-NPU 协调之间的预填充速度和端到端延迟比较。

每秒令牌数。当所有异常值都被修剪时，生成精度达到最低点，而生成速度达到最高点。具体来说，Qwen1.5-1.8B模型的准确率仅为8.1%，生成速度为596个token/s；Gemma-2B模型的准确度为41.9%，生成速度为345个令牌/秒。

4.5 内存消耗

我们将llm.npu与INT8权重基线进行比较，因为移动NPU仅支持INT8权重计算。图17显示了使用512令牌提示的Redmi K60 Pro上的内存消耗结果。llm.npu比llama.cpp和TFLite多消耗 $1.32 \times$ 的内存。开销是由于MLLM和QNN框架造成的，它们为每个运算符分配独立的激活缓冲区以提高速度。llm.npu引入的微小额外内存开销是其§3.3阴影离群值执行技术（黑色），该技术将微小的浮点权重加载到内存中，仅占总内存的0.6%–1%。

4.6 GPU-NPU协调分析

llm.npu与移动GPU兼容；不过，由于实现简单，我们目前只实现了CPU-NPU版本。为了说明使用移动GPU的潜在优势，我们模拟GPU-NPU协同调度，假设TFLite作为GPU后端。使用Gemma-2B模型在不同的提示长度和Longbench数据集上进行评估，特别是在Redmi K70 Pro上，如图18所示。我们的观察表明，虽然GPU-NPU协调

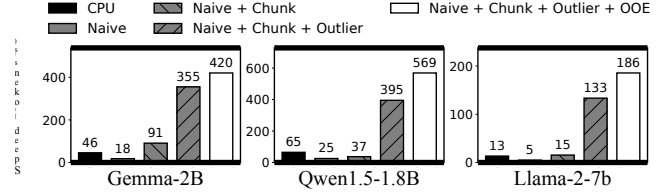


图 19. llm.npu 的消融研究（提示长度=512）。

不提升预填充速度，有效降低端到端时延。

如图18(a)所示，通过GPU-NPU协调实现的预填充速度与CPU-NPU协调的预填充速度相当。出现这种现象的原因是，在llm.npu的设计中，对移动CPU和GPU的计算需求明显低于对NPU的计算需求（详见3.4节）。因此，移动CPU和GPU的计算延迟可以被NPU计算有效隐藏，这表明移动CPU或移动GPU的选择并不重要。另一方面，与CPU-NPU协调相比，GPU-NPU协调可以将端到端延迟减少80-90毫秒。这一改进归功于GPU相对于CPU更快的解码速度。

4.7 消融研究。

我们对使用Qwen1.5-1.8B、Gemma-2B和LlaMA2-7B模型的llm.npu每种技术带来的好处进行了全面的细分分析，如图19所示。最左边的条代表使用llama.cpp在CPU上的预填充速度。第二个条显示了移动NPU上的简单实现，然后是我们三种技术的结合，最右边的条代表llm.npu。这三种技术分别用chunk (§3.2)、outlier (§3.3)和OOE (§3.4)表示。我们观察到llm.npu的三种技术对整体改进做出了重大贡献。

首先，由于LLM和移动NPU之间存在巨大差距，直接将LLM预填充工作负载卸载到移动NPU会导致 2.55 – $2.68 \times$ 延迟，如§2.3中详述。此外，块共享图通过减少图构建和优化延迟，将预填充速度提高了 1.46 – $5.09 \times$ 。Gemma-2B模型实现了最高的加速，因为它需要更多的时间来构建和优化。此外，阴影异常值执行将预填充延迟减少了 3.91 – $8.68 \times$ ，允许每个张量MatMul操作以最小的CPU/GPU开销充分利用NPU。最后，无序子图执行通过减少NPU内的执行气泡，将预填充延迟减少了18%–44%。

5 讨论和未来的工作

NPU相对于GPU的优势。通常，移动GPU和NPU都充当设备上的硬件加速器

DNN inference [53, 86]. However, as mobile GPUs are already extensively used for graphic rendering on devices, relying on them for LLM inference can cause significant resource contention. In contrast, mobile NPUs, which are dedicated exclusively to deep learning tasks, have relatively lighter workloads, leaving ample time to run LLM inference. Besides, as elaborated in Section 2.2, mobile NPUs provide greater computational capacity, enhanced energy efficiency, and increased availability compared to mobile GPUs, making them more suitable for on-device LLM inference.

11m.npu’s technique stability. Despite the recent trend towards supporting FP16 in advanced NPUs [6, 13, 17], 11m.npu remains essential for NPU-based prefilling acceleration for two reasons: (1) Integer operations are expected to maintain dominance in mobile NPU-side LLM inference due to their superior computation performance, memory efficiency, and energy savings, while CPU/GPUs serve as supplementary units for enhancing accuracy. Therefore, 11m.npu’s three INT8-related techniques, which include co-scheduling across CPU/GPU and NPU to achieve high accuracy and efficiency, will continue to have a significant impact in this domain. (2) 11m.npu is capable of empowering legacy devices with on-device LLM capabilities, democratizing access to LLMs.

Future hardware design implications. 11m.npu’s performance could be further enhanced with the following hardware optimizations: (1) *Dynamic shape-aware optimization*: Hardware capable of dynamically reconfiguring to accommodate varying input and output shapes would markedly improve the acceleration of LLM inference. (2) *Increased data cache size*: Mobile NPUs generally enhance inference performance by loading an entire layer’s weights into the data cache. Increasing the NPU cache size to align with LLM weight sizes is crucial for optimal performance. (3) *Mixed-precision operands in computing units*: Mobile NPUs typically support only uniform precision operands, necessitating additional data format conversions to perform FP16 * INT8 operations. This overhead can be eliminated by enabling support for mixed-precision operands in computing units.

11m.npu with mobile GPUs. Our current implementation relies solely on mobile CPUs for floating-point operators and the decoding phase due to ease of implementation. However, the design of 11m.npu is compatible with both mobile CPUs and GPUs. We plan to incorporate GPU-NPU coordination in future developments. The primary engineering efforts in integrating mobile GPUs into 11m.npu involve configuring the GPU backend and implementing the necessary operators. With mobile GPUs, 11m.npu can significantly reduce end-to-end inference latency, as elaborated in Section 4.6.

6 Related work

On-device LLM optimization. To reduce the substantial memory consumption of on-device LLM inference, various

compression techniques have been proposed, including quantization and knowledge distillation [30, 35, 38, 46, 58, 65, 70, 80, 88, 91, 92, 96, 98, 101]. To minimize on-device LLM computation, researchers have introduced *token pruning* [29, 31, 51, 72, 79], which prunes unnecessary tokens during the inference process. Speculative decoding, a method that accelerates token generation by offloading tasks to a smaller LLM, has been widely adopted in open-source frameworks [3, 59] and extensively researched [32, 36, 42, 50, 61, 87, 95]. As a system optimization, 11m.npu is orthogonal and compatible with these algorithm-level optimizations.

On-chip offloading for ML. This has been thoroughly studied to enable faster DNN inference by leveraging heterogeneous mobile processors like GPUs and NPUs [37, 39, 41, 45, 52–54, 63, 66, 85, 86, 89, 93, 94, 104, 105]. MobiSR [54] utilizes mobile NPUs to speed up super-resolution computation. However, these methods do not address LLM-specific features and are unsuitable for on-device LLM scenarios. The most relevant work is PowerInfer-V2 [94], which also utilizes mobile NPUs for the prefilling. But, it does not comprehensively analyze the challenges of on-device NPU offloading for LLM prefilling or incorporate techniques to fully harness NPU capability. Our experiments demonstrate that our approach is up to 5.32× more efficient in the prefill stage. 11m.npu is inspired by these efforts and is the first LLM inference framework with efficient on-device NPU offloading.

Mobile NPU execution optimization. With the growing importance of mobile NPUs in smartphones, significant efforts have been made to optimize their execution efficiency [34, 64, 71, 73, 76, 78]. Pitchfork [73] defines a portable fixed-point intermediate representation to optimize fixed-point execution efficiency. Isaria [76] proposes a framework for automatically generating vectorizing compilers for DSP architectures, creating efficient operator codes for mobile NPUs. As a system framework, 11m.npu is orthogonal and can leverage them to generate more efficient operator libraries as execution backend, further boosting performance.

7 Conclusions

This paper has proposed 11m.npu, the first LLM inference system utilizing on-device NPU offloading to reduce prefill latency and energy consumption. 11m.npu has incorporated novel techniques: chunk-sharing graph, shadow outlier execution and out-of-order subgraph execution to enhance NPU offloading efficiency. Extensive experiments have demonstrated 11m.npu to show its superior performance benefits, e.g, up to 43.6× speedup and 59.5× energy savings.

8 Acknowledgement

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DNN 推理 [53, 86]。然而，由于移动 GPU 已经广泛用于设备上的图形渲染，依赖它们进行 LLM 推理可能会导致严重的资源争用。相比之下，专门用于深度学习任务的移动 NPU 的工作负载相对较轻，因此有充足的时间来运行 LLM 推理。此外，如第 2.2 节所述，与移动 GPU 相比，移动 NPU 提供了更大的计算能力、更高的能源效率和更高的可用性，使它们更适合设备上的 LLM 推理。

llm.npu 的技术稳定性。尽管最近的趋势是在高级 NPU 中支持 FP16 [6,13,17]，但 llm.npu 对于基于 NPU 的预填充加速仍然至关重要，原因有两个：（1）整数运算由于其卓越的计算性能、内存效率和节能而有望在移动 NPU 端 LLM 推理中保持主导地位，而 CPU/GPU 作为补充单元来提高准确性。因此，llm.npu 的三种 INT8 相关技术，包括跨 CPU/GPU 和 NPU 的协同调度以实现高精度和高效率，将继续对该领域产生重大影响。（2）llm.npu 能够为传统设备提供设备上的 LLM 功能，从而实现 LLM 访问的民主化。

未来硬件设计的影响。llm.npu 的性能可以通过以下硬件优化得到进一步增强：（1）动态形状感知优化：能够动态重新配置以适应不同的输入和输出形状的硬件将显著提高 LLM 推理的加速。（2）增加数据缓存大小：移动 NPU 通常通过将整个层的权重加载到数据缓存中来增强推理性能。增加 NPU 缓存大小以与 LLM 权重大小保持一致对于获得最佳性能至关重要。（3）计算单元中的混合精度操作数：移动 NPU 通常仅支持统一精度操作数，需要额外的数据格式转换来执行 FP16 * INT8 运算。通过在计算单元中启用对混合精度操作数的支持可以消除这种开销。

带有移动 GPU 的 llm.npu。由于易于实现，我们当前的实现仅依赖于移动 CPU 进行浮点运算符和解码阶段。不过，llm.npu 的设计同时兼容移动 CPU 和 GPU。我们计划在未来的开发中纳入 GPU-NPU 协调。将移动 GPU 集成到 llm.npu 的主要工程工作包括配置 GPU 后端和实现必要的运算符。借助移动 GPU，llm.npu 可以显著减少端到端推理延迟，如第 4.6 节所述。

6 相关工作

设备上 LLM 优化。为了减少设备上 LLM 推理的大量内存消耗，各种

已经提出了压缩技术，包括量化和知识蒸馏[30,35,38,46,58,65,70,80,88,91,92,96,98,101]。为了最大限度地减少设备上的 LLM 计算，研究人员引入了令牌修剪 [29,31,51,72,79]，它在推理过程中修剪不必要的令牌。推测解码是一种通过将任务卸载到较小的 LLM 来加速令牌生成的方法，已在开源框架中广泛采用[3, 59]，并进行了广泛的研究[32, 36, 42, 50, 61, 87, 95]。作为系统优化，llm.npu 是正交的并且兼容这些算法级别的优化。

ML 的片上卸载。人们已经对此进行了深入研究，以便通过利用 GPU 和 NPU 等异构移动处理器来实现更快的 DNN 推理 [37, 39, 41, 45, 52–54, 63, 66, 85, 86, 89, 93, 94, 104, 105]。MobiSR [54]利用移动 NPU 来加速超分辨率计算。然而，这些方法并没有解决 LLM 特定的功能，并且不适合设备上的 LLM 场景。最相关的工作是 PowerInfer-V2 [94]，它也利用移动 NPU 进行预填充。但是，它没有全面分析 LLM 预填充的设备上 NPU 卸载的挑战，也没有结合技术来充分利用 NPU 功能。我们的实验表明，我们的方法在预填充阶段的效率提高了 5.32×。llm.npu 受到这些努力的启发，是第一个具有高效设备上 NPU 卸载的 LLM 推理框架。移动 NPU 执行优化。随着移动 NPU 在智能手机中的重要性日益增加，人们付出了巨大的努力来优化其执行效率 [34,64,71,73,76,78]。Pitchfork [73]定义了一种可移植的定点中间表示来优化定点执行效率。Isaria [76]提出了一个框架，用于自动生成 DSP 架构的矢量化编译器，为移动 NPU 创建高效的运算符代码。作为一个系统框架，llm.npu 是正交的，可以利用它们生成更高效的算子库作为执行后端，进一步提高性能。

7 结论

本文提出了 llm.npu，这是第一个利用设备上 NPU 卸载来减少预填充延迟和能耗的 LLM 推理系统。llm.npu 融入了新技术：块共享图、影子离群值执行和乱序子图执行，以提高 NPU 卸载效率。大量实验证明 llm.npu 具有卓越的性能优势，例如高达 43.6× 的加速和 59.5× 的节能。

8 致谢

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A Artifact Appendix

A.1 Abstract

The artificial evaluations primarily present the prefill performance results detailed in Section 4.2 and the inference accuracy evaluation results discussed in Section 4.4. These results can be found in the *performance_results* and *accuracy_results* directories in our GitHub repository¹. To conduct these evaluations, please refer to the README.md file in the repository. Each evaluation directory in the repository contains detailed instructions for installing and executing evaluations for *llm.npu* and all baseline methods. This includes guidance on environment setup, LLM and dataset downloads, source code compilation and execution, and clarity on results. We recommend performing the accuracy evaluation first, followed by the prefill performance test. For optimal results, use a Redmi K70 Pro 24G for on-device LLM inference and an NVIDIA A100 40G GPU server for testing, as these setups have been validated by the authors. Most experiments utilize Qwen1.5-1.8B as a representative example to demonstrate the functionality and evaluation results of *llm.npu*. Users have the flexibility to tailor their own LLM models by adjusting the code, if needed. Once this artifact evaluation is accepted, we will archive the evaluation results.

A.2 Artifact check-list (meta-information)

- **Compilation: Support compilation from users.**
- **Binary: Compiled binaries are available.**
- **Model: Qwen1.5-1.8B-Chat is used as demonstration model. Gemma-2B, Phi-2-2.7B, LLaMA-2-7B, and Mistral-7B are also supported.**
- **Dataset: HellaSwag [103] and LAMBADA [68] are used as demonstration datasets. OpenBookQA [62], WinoGrande [4], and MMLU [43] are also supported.**
- **Server Hardware: NVIDIA-A100 40G**
- **Device Hardware: Redmi K70 Pro 24G**
- **Execution: C++, Python and bash scripts**
- **Metrics: inference accuracy and prefill speed**
- **Output: log in terminal**
- **Experiments: Section 4.2 and Section 4.4**
- **How much disk space required (approximately)?: 200GB**
- **How much time is needed to prepare workflow (approximately)?: 5 hours**
- **How much time is needed to complete experiments (approximately)?: 2 hours**
- **Publicly available?: Yes**
- **Code licenses (if publicly available)?: MIT**
- **Data licenses (if publicly available)?: MIT**
- **Workflow automation framework used?: mllm [23]**
- **Archived (provide DOI)?: <https://doi.org/10.5281/zenodo.14392760>.**

A.3 Description

A.3.1 How to access. Our artificial evaluation is publicly accessible at

¹<https://doi.org/10.5281/zenodo.14392760>

<https://doi.org/10.5281/zenodo.14392760>.

A.3.2 Hardware dependencies. For the accuracy results discussed in Section 4.4, reviewers will require access to a GPU server equipped with an NVIDIA A100 40GB GPU, x86 CPU cores, and preferably over 200GB of CPU memory. The server should operate on Ubuntu 20.04 OS.

To assess the performance results in Section 4.2, reviewers will need a Redmi K70 Pro with 24GB of memory.

A.3.3 Software dependencies. For the accuracy results outlined in Section 4.4, the server should have CUDA 12.5 installed and a Conda virtual environment with Python version 3.8 or higher.

A.3.4 Datasets. The datasets used for accuracy results include HellaSwag [103], LAMBADA, OpenBookQA [62], WinoGrande [4], and MMLU [43]. All datasets can be automatically downloaded using our code from Hugging Face [21].

A.3.5 Models. We demonstrate *llm.npu*'s functionality on devices using Qwen1.5-1.8B. Users may also perform additional evaluations by integrating customized models like Gemma-2B, Phi-2-2.7B, LLaMA-2-7B, and Mistral-7B.

A.4 Installation

To obtain the accuracy results discussed in Section 4.4, users should begin by using Conda to install the Python virtual environment specified in *accuracy_results/environment.yml*. This environment includes all the necessary dependencies to reproduce the accuracy results of *llm.npu*. When evaluating the baseline methods, particularly SmoothQuant [84], users need to install the SmoothQuant environment as detailed in *accuracy_results/baselines/README.md*.

For the performance results outlined in Section 4.2, users are advised to refer to and follow the installation instructions provided in each respective *README.md* file within the *performance_results* directory.

A.5 Experiment workflow

Initially, users should perform the accuracy results evaluation (*accuracy_results*) to generate the quantized model necessary for on-device inference with *llm.npu* (*accuracy_results/llm.npu/examples/act_scales_try/llmnpn_get_int8_weights_finalmodel_int8bias_ns.py*). Subsequently, using the generated model, users should proceed with the performance results evaluation (*performance_results*).

A.6 Evaluation and expected results

We offer two primary types of evaluation results: accuracy evaluation results for Section 4.4 and prefill performance evaluation results for Section 4.2. These results contribute to the data reported in Table 6 and illustrated in Figure 14.

A.7 Methodology

Availability, Functional, and Reproducibility

神器附录

A.1 摘要

人工评估主要呈现第 4.2 节中详述的预填充性能结果和第 4.4 节中讨论的推理准确性评估结果。这些结果可以在我们的 GitHub 存储库¹ 的 Performance_results 和 Accuracy_results 目录中找到。要进行这些评估，请参阅存储库中的 README.md 文件。存储库中的每个评估目录都包含有关安装和执行 llm.npu 和所有基线方法评估的详细说明。这包括环境设置、LLM 和数据集下载、源代码编译和执行以及结果清晰度的指导。我们建议首先进行精度评估，然后进行预填充性能测试。为了获得最佳结果，请使用 Redmi K70 Pro 24G 进行设备上 LLM 推理，并使用 NVIDIA A100 40G GPU 服务器进行测试，因为这些设置已经过作者的验证。大多数实验都以 Qwen1.5-1.8B 为代表示例来演示 llm.npu 的功能和评估结果。如果需要，用户可以通过调整代码灵活地定制自己的法学硕士模型。一旦工件评估被接受，我们将存档评估结果。

A.2 工件检查表（元信息）

- 编译：支持用户编译。
- 二进制：已编译的二进制文件可用。
- 示范模型采用型号：Qwen1.5-1.8B-Chat。还支持 Gemma-2B、Phi-2-2.7B、LLaMA-2-7B 和 Mistral-7B。
- 数据集：HellaSwag [103] 和 LAMBADA [68] 用作演示数据集。还支持 OpenBookQA [62]、WinoGrande [4] 和 MMLU [43]。
- 服务器硬件：NVIDIA-A100 40G
- 设备硬件：Redmi K70 Pro 24G
- 执行：C++、Python 和 bash 脚本
- 指标：推理准确性和预填充速度
- 输出：登录终端
- 实验：第 4.2 节和第 4.4 节
- 需要多少磁盘空间（大约）：200GB
- 准备工作流程需要多长时间（大约）？：5 小时
- 完成实验需要多长时间（大约）：2 小时
- 是否公开？：是
- 代码许可证（如果公开可用）？：麻省理工学院
- 数据许可证（如果公开可用）？：麻省理工学院
- 使用的工作流自动化框架？：mllm [23]
- 已存档（提供 DOI）？：https://doi.org/10.5281/zenodo.14392760。

A.3 描述

A.3.1 如何访问。我们的人工评估可公开访问

<https://doi.org/10.5281/zenodo.14392760>。

A.3.2 硬件依赖性。对于第 4.4 节中讨论的准确性结果，审核者将需要访问配备 NVIDIA A100 40GB GPU、x86 CPU 内核以及最好超过 200GB CPU 内存的 GPU 服务器。服务器应在 Ubuntu 20.04 操作系统上运行。

为了评估第 4.2 节中的性能结果，评测人员需要配备 24GB 内存的 Redmi K70 Pro。

A.3.3 软件依赖性。为了获得第 4.4 节中概述的准确性结果，服务器应安装 CUDA 12.5 和具有 Python 3.8 或更高版本的 Conda 虚拟环境。

A.3.4 数据集。用于准确性结果的数据集包括 HellaSwag [103]、LAMBADA、OpenBookQA [62]、WinoGrande [4] 和 MMLU [43]。所有数据集都可以使用我们的 Hugging Face [21] 代码自动下载。

A.3.5 模型。我们在使用 Qwen1.5-1.8B 的设备上演示 llm.npu 的功能。用户还可以通过集成 Gemma-2B、Phi-2-2.7B、LLaMA-2-7B 和 Mistral-7B 等定制模型来执行其他评估。

A.4 安装

为了获得第 4.4 节中讨论的准确性结果，用户应该首先使用 Conda 安装在 precision_results/environment.yml 中指定的 Python 虚拟环境。该环境包含重现 llm.npu 准确性结果所需的所有依赖项。在评估基线方法时，特别是 SmoothQuant [84]，用户需要安装 SmoothQuant 环境，详细信息请参阅 precision_results/baselines/README.md。

对于第 4.2 节中概述的性能结果，建议用户参考并遵循 Performance_results 目录中每个相应 README.md 文件中提供的安装说明。

A.5 实验流程

最初，用户应该执行精度结果评估（accuracy_results），以生成使用 llm.npu 进行设备上推理所需的量化模型（accuracy_results/llm.npu/examples/act_scales_try/llmnp_get_int8_weights_finalmodel_int8bias_ns.py）。随后，使用生成的模型，用户应继续进行性能结果评估（performance_results）。

A.6 评价和预期结果

我们提供两种主要类型的评估结果：第 4.4 节的准确性评估结果和第 4.2 节的预填充性能评估结果。这些结果构成了表 6 中报告的数据和图 14 中所示的数据。

A.7 方法论

可用性、功能性和可重复性

¹<https://doi.org/10.5281/zenodo.14392760>

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